

Video Info

File name: P-CDL-I-M4-1-Modernize Infrastructure and Applications with Google Cloud

Presenter:



04

Modernize Infrastructure and Applications with Google Cloud

Module 4

Modernize Infrastructure and Applications with Google Cloud



Lessons

- 01 Important cloud migration terms
- 02 Modernizing infrastructure in the cloud
- 03 Modernizing applications in the cloud

Module 4

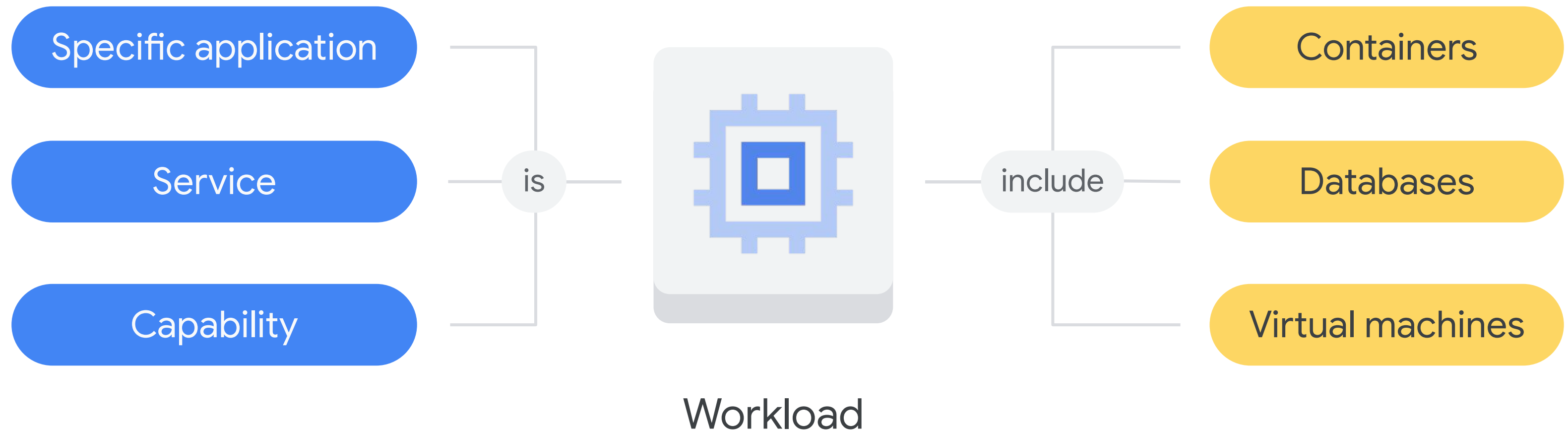
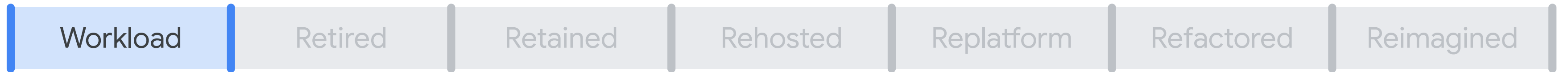
Modernize Infrastructure and Applications with Google Cloud



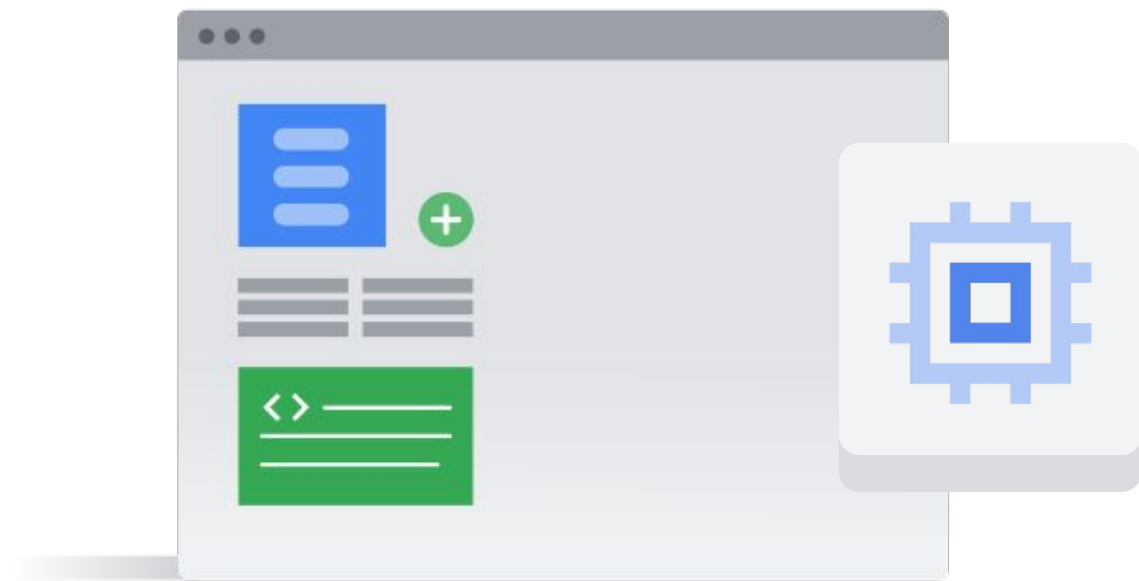
Lessons

- 01 Important cloud migration terms
- 02 Modernizing infrastructure in the cloud
- 03 Modernizing applications in the cloud

Important cloud migration terms



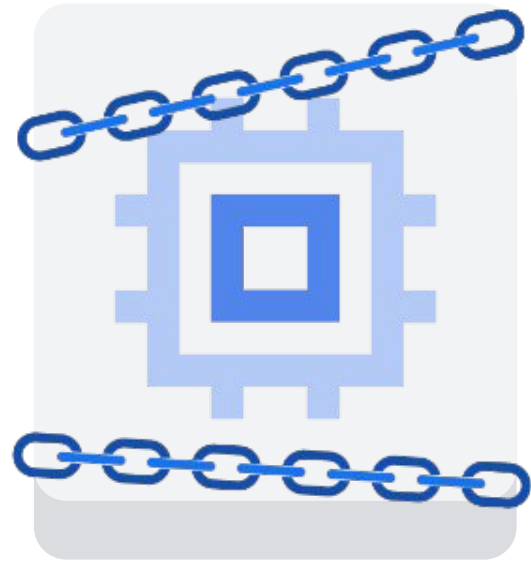
Important cloud migration terms



Retiring a workload means removing it from a platform.

- Unnecessary
- Not cost-effective
- Not secure
- Not compatible with a specific platform

Important cloud migration terms



Intentionally
keeping a workload

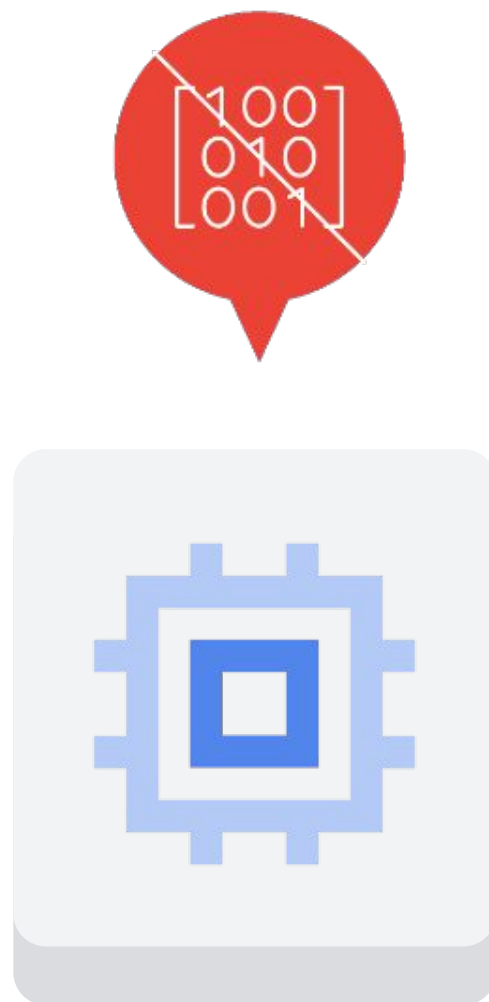
Kept on-premises

Kept in a hybrid cloud environment

Managed by the business

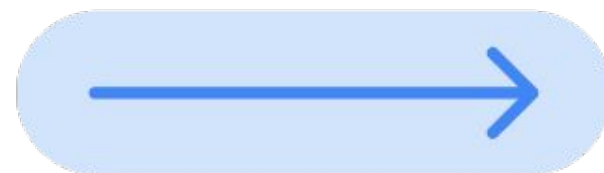
Not subject to the same level of cloud provider control

Important cloud migration terms

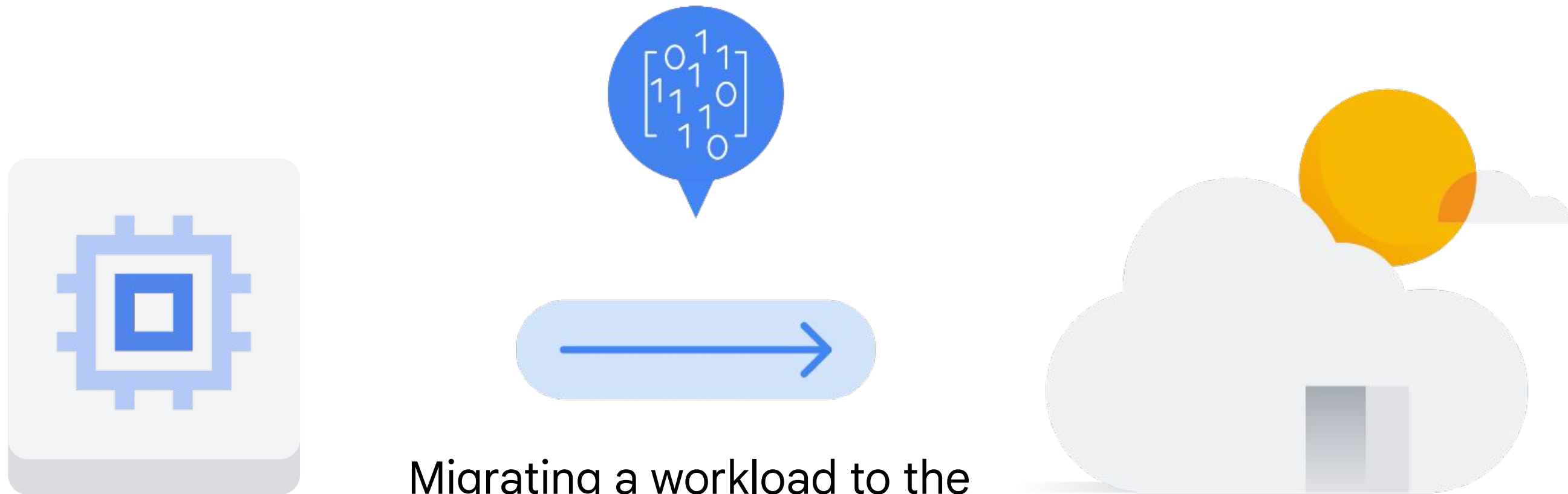


Workload

Migration of a workload to the cloud without changing anything in the workload's code or architecture



Important cloud migration terms



Workload

Migrating a workload to the cloud while making some changes to the workload's code or architecture

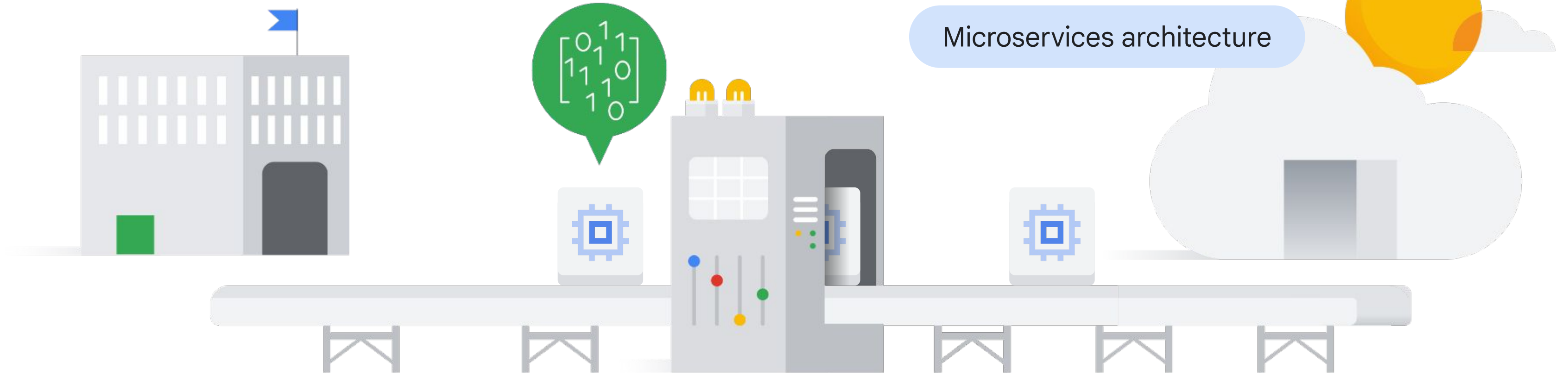
Important cloud migration terms



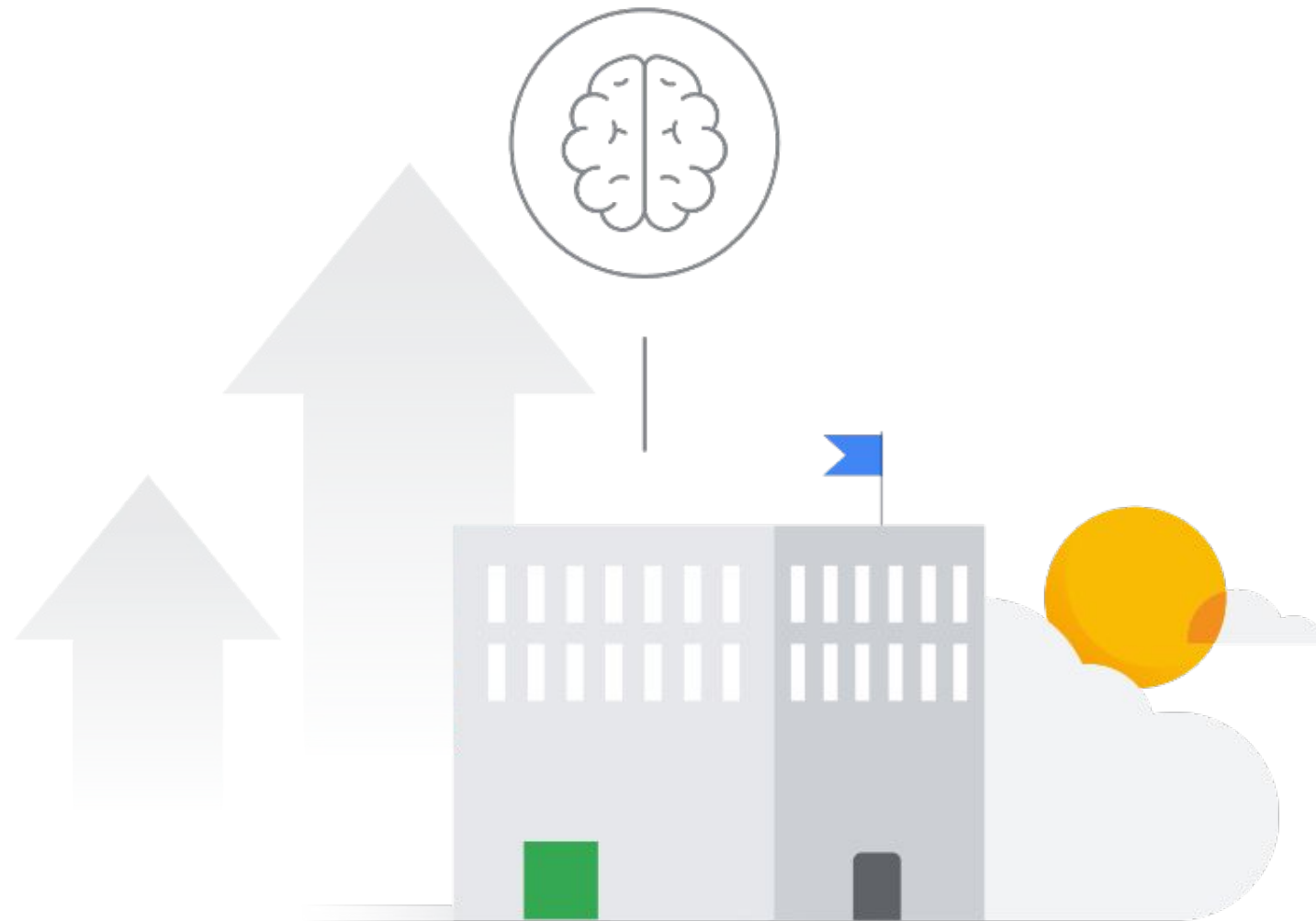
Changing the
code of a workload

Serverless architecture

Microservices architecture



Important cloud migration terms



The process of rethinking how an organization uses technology to achieve its business goals

Cloud strategy

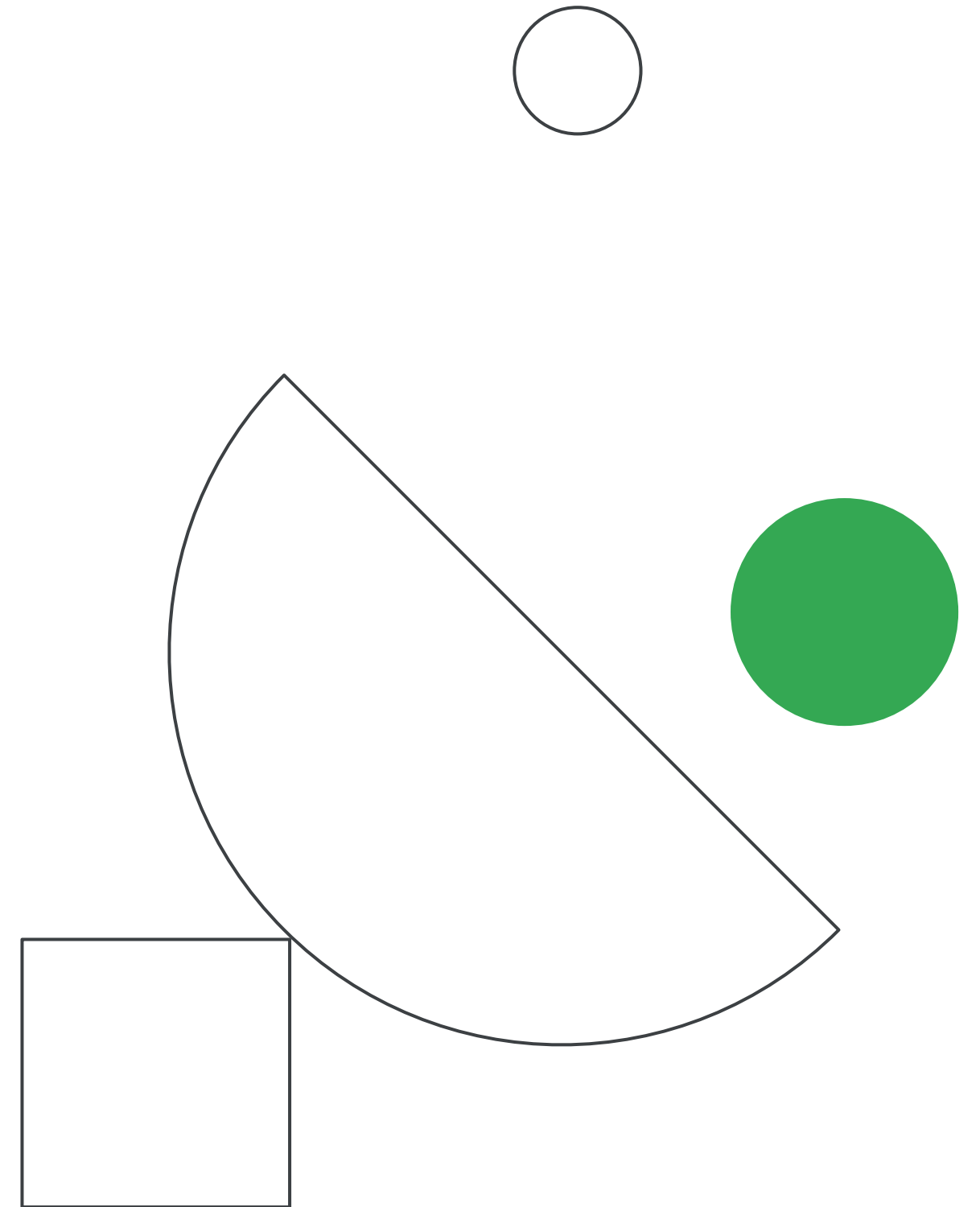
Artificial intelligence

Machine learning

Scenario 1

A multinational coffeehouse chain currently accepts orders in-person, by phone, or on their website. They want to provide customers with another way to order by building a new smartphone app. The app needs to be scalable and to function globally at low latency.

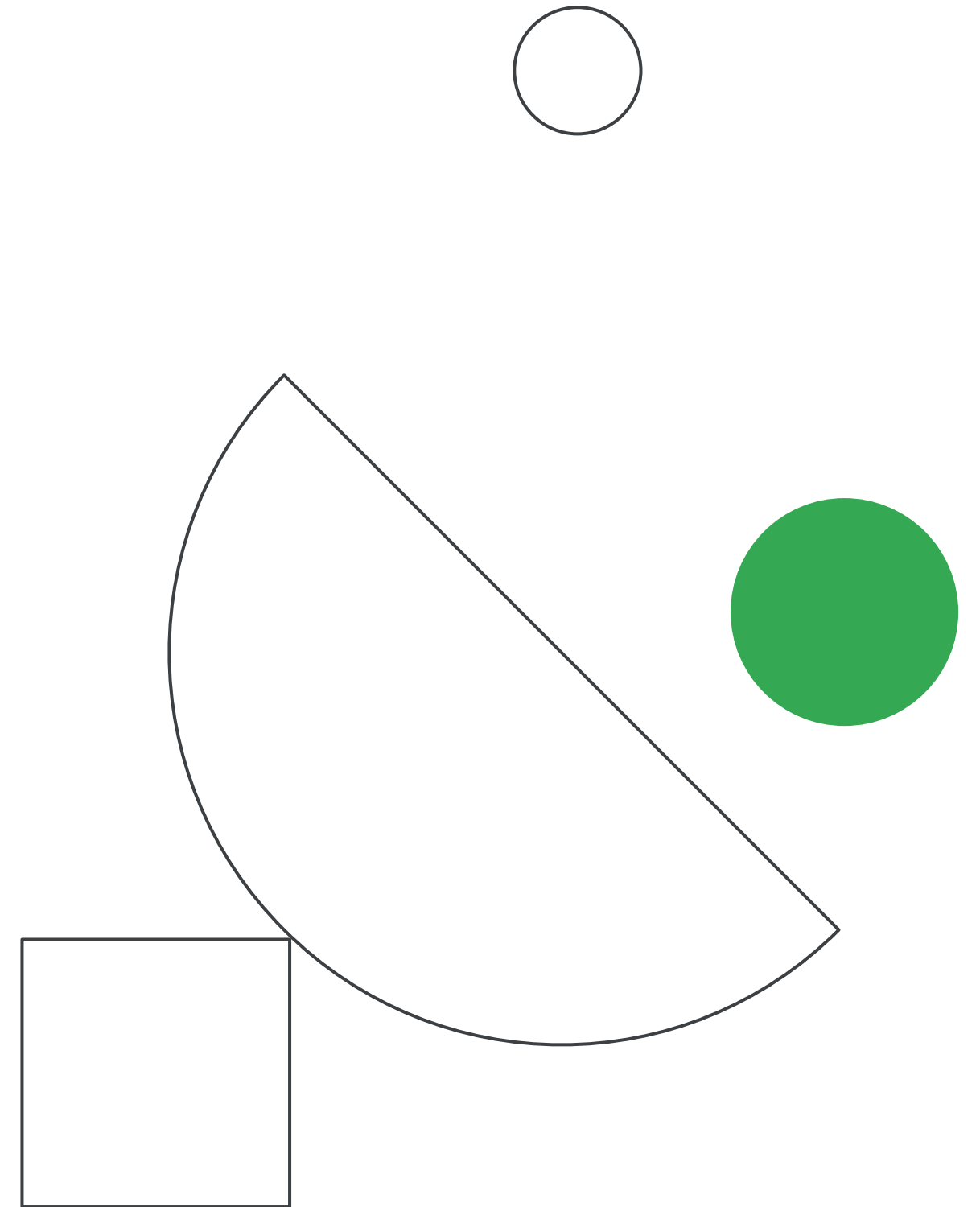
Which modernization strategy should they adopt?



Scenario 1

Answer: Reimagine

Why? The organization wants to develop a new mobile application, rather than updating an existing product. To ensure the app functions as needed it should be developed in the cloud.

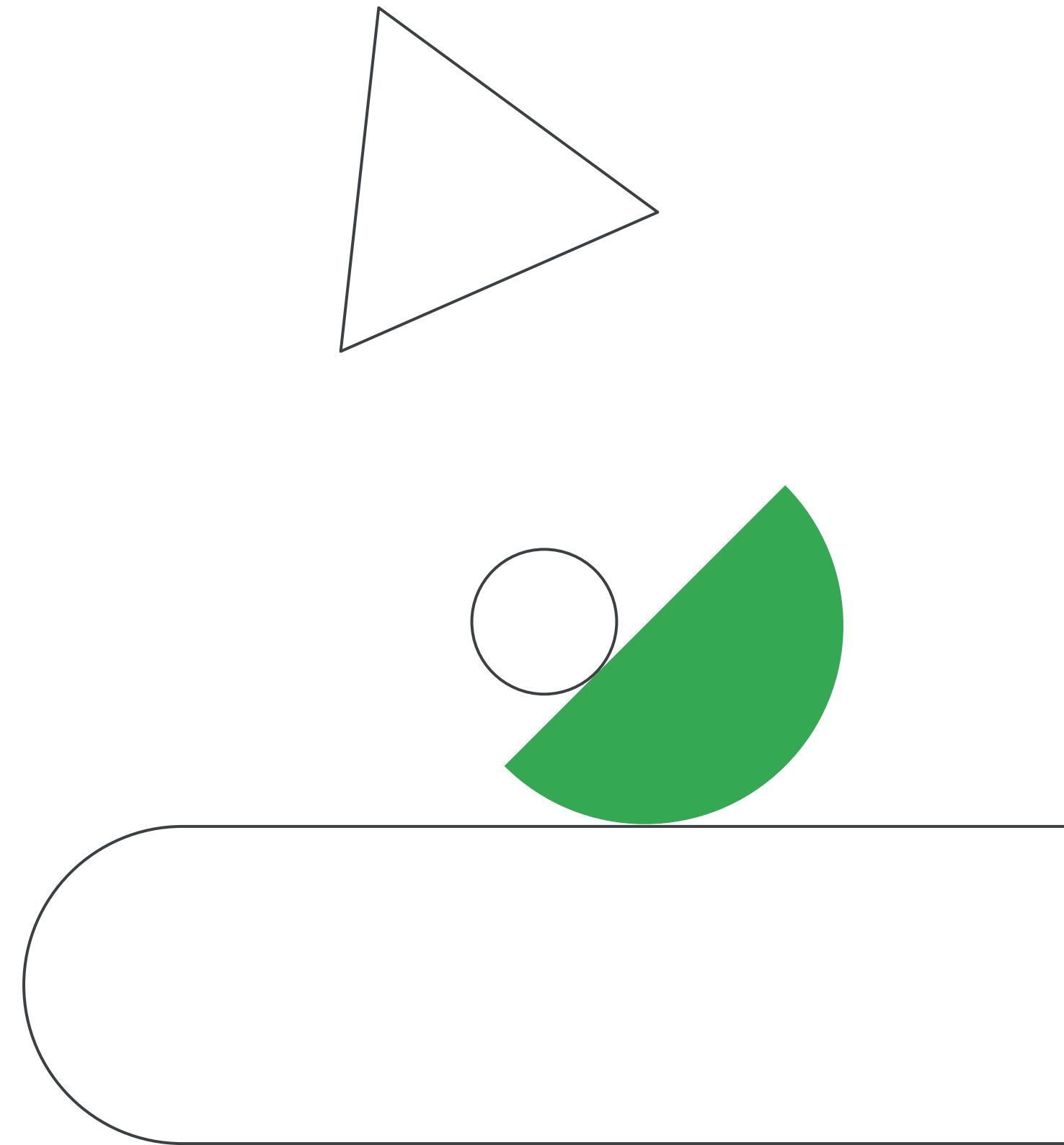


Scenario 2

A video game manufacturer distributes its products through applications on consoles and mobile devices, supported by legacy infrastructure.

Consumers are complaining that the app lacks features like cloud gaming and customizable profiles. The manufacturer wants to provide these, but is struggling to integrate them into the current application and can't risk extended downtime.

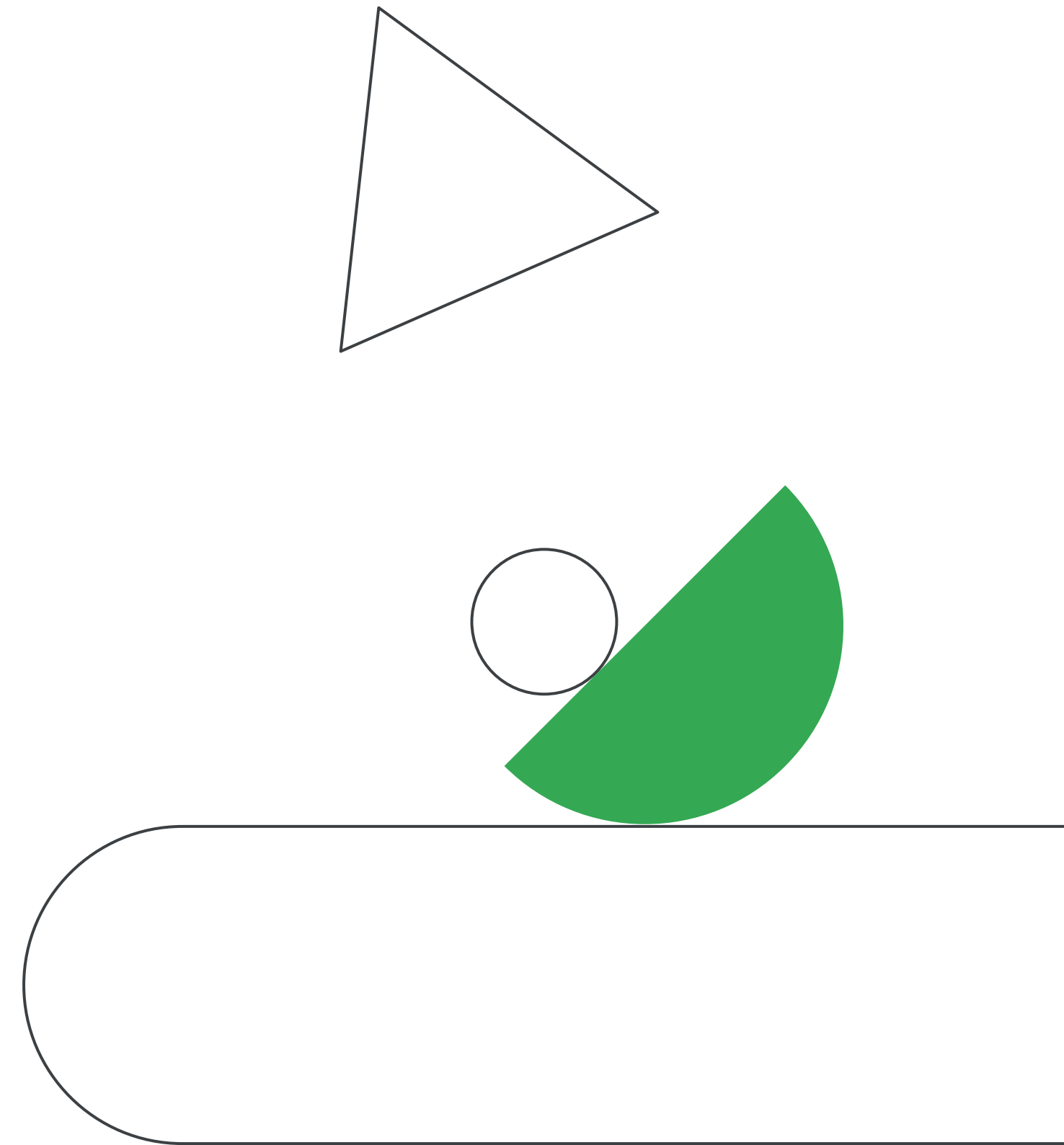
Which modernization strategy should they adopt?



Scenario 2

Answer: Refactor

Why? Upgrading the existing application with these new features would be difficult due to legacy hardware limitations and the costs of downtime. Refactoring will allow them to integrate the new features into the existing application without having to risk extended downtime.

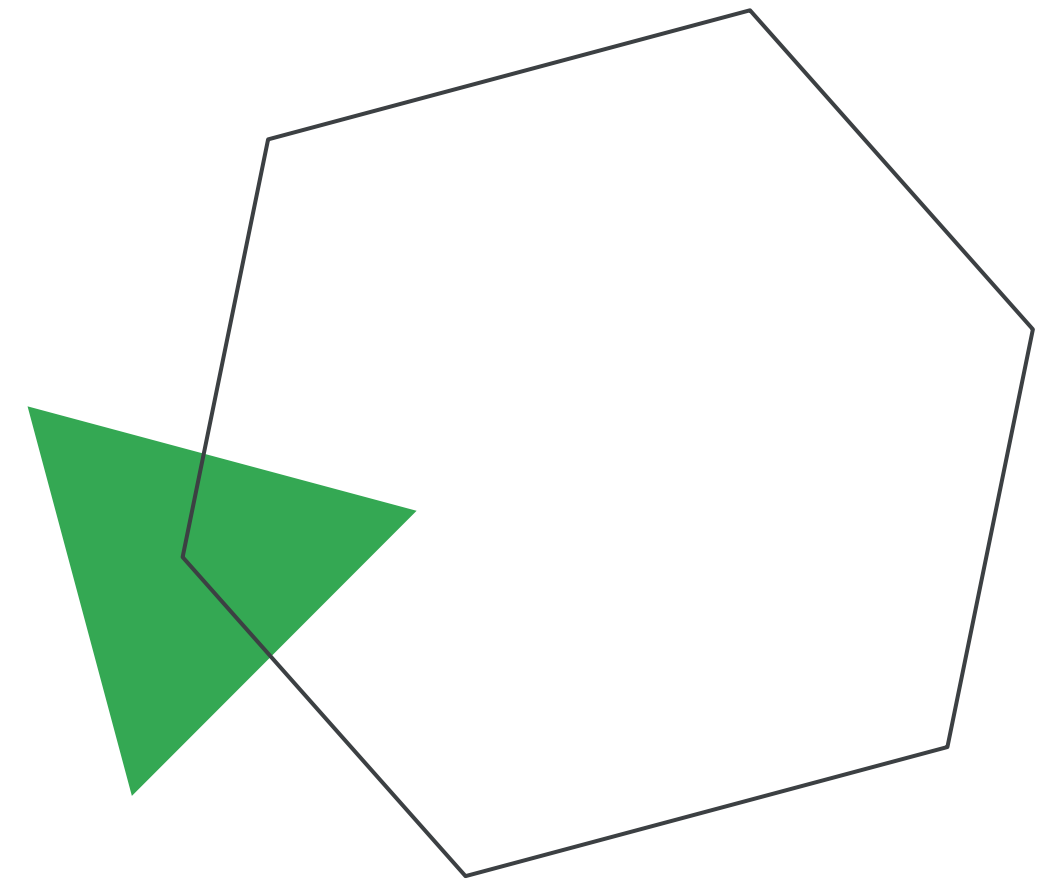
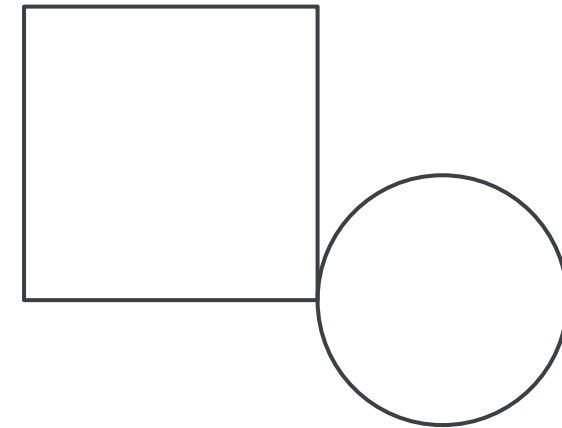


Scenario 3

A financial services company has a legacy application that is used to process customer transactions. The application is running on an outdated operating system and is not compatible with the company's newer hardware.

The financial services company wants to modernize the application without having to rewrite the entire application from the beginning.

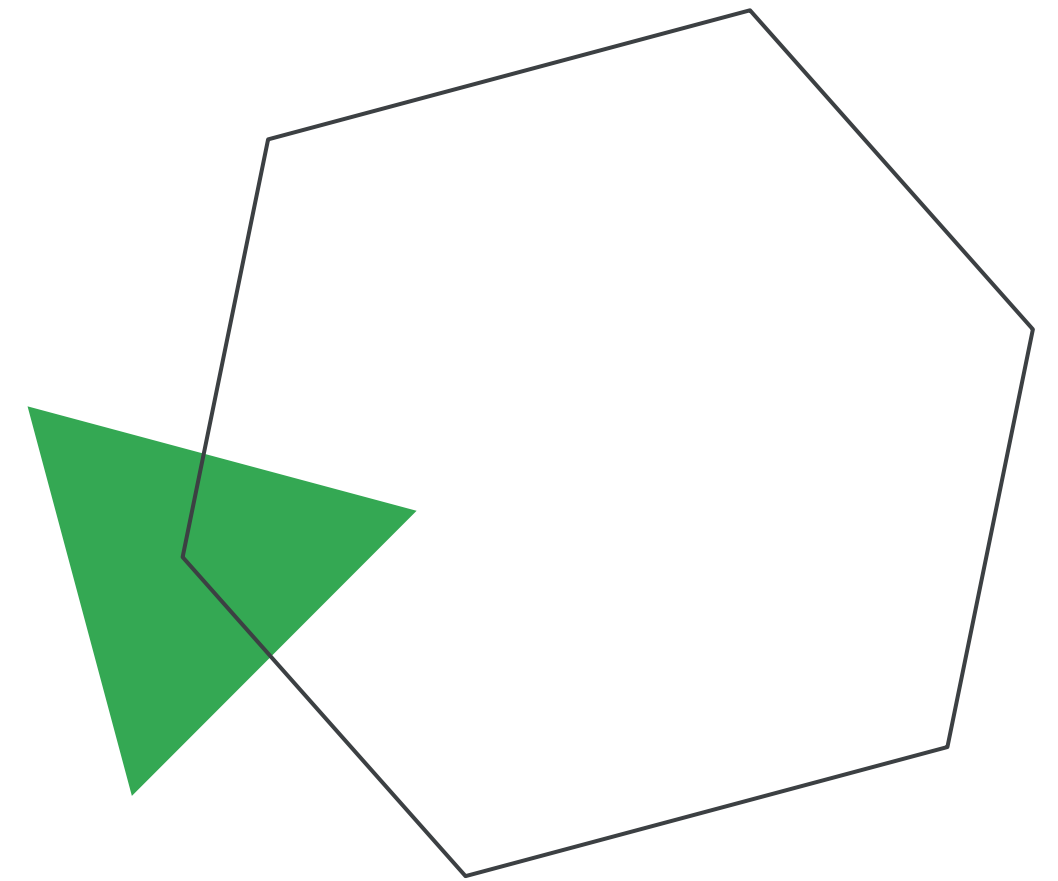
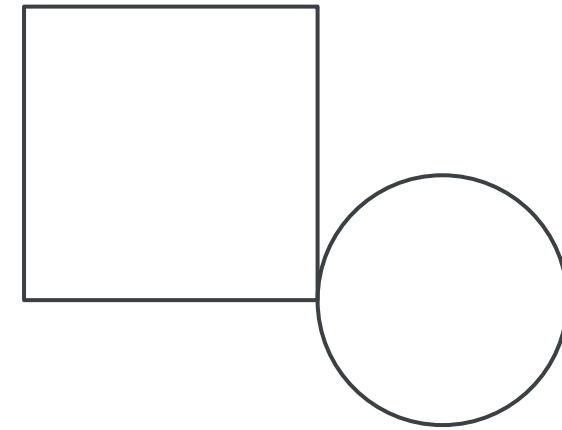
Which modernization strategy should they adopt?



Scenario 3

Answer: Replatform

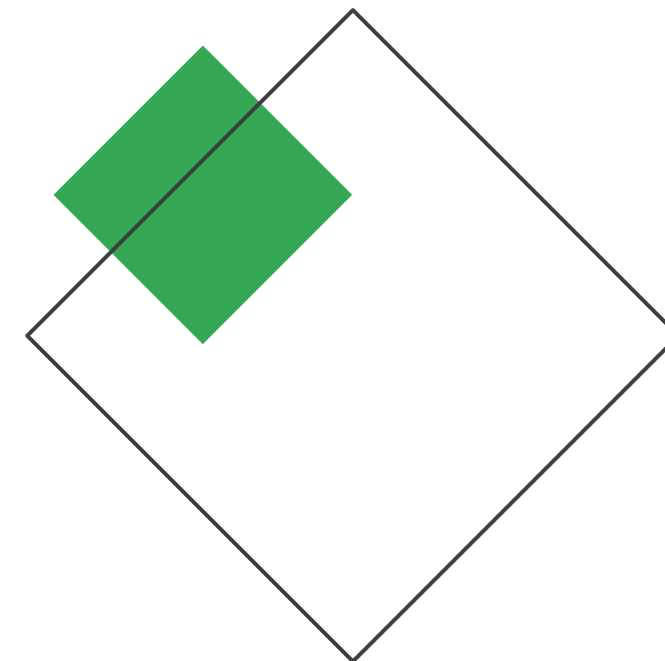
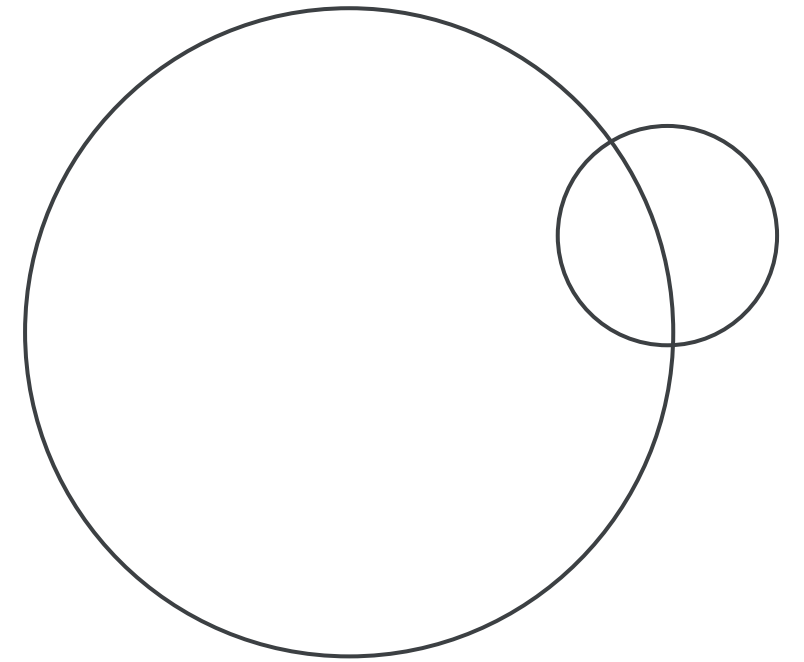
Why? The replatform strategy involves making minor changes to the application code to take advantage of the new operating system's features. This will allow the financial services company to modernize the application without having to rewrite the entire application from the beginning.



Scenario 4

A high-profile law firm has a large on-premises database of legal resources which can be accessed on-site, or on a web portal which is slow and unreliable. The firm now wants its databases to be accessible quickly and easily from anywhere.

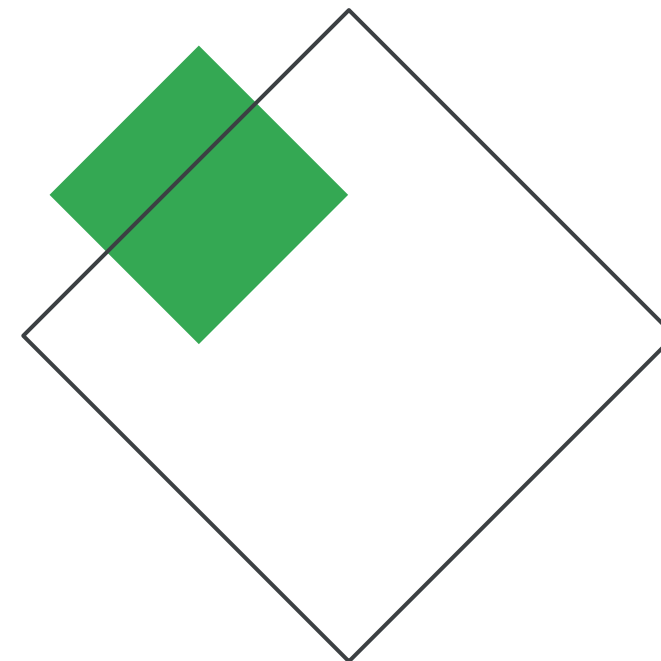
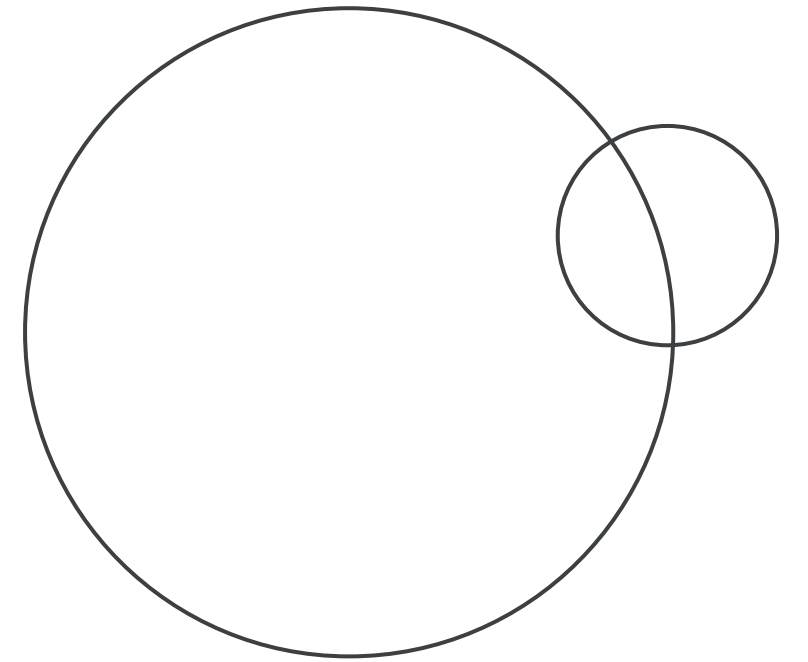
Which modernization strategy should they adopt?



Scenario 4

Answer: **Rehost**

Why? Application redevelopment is not required here, as the migration of databases to the cloud is relatively simple.



Module 4

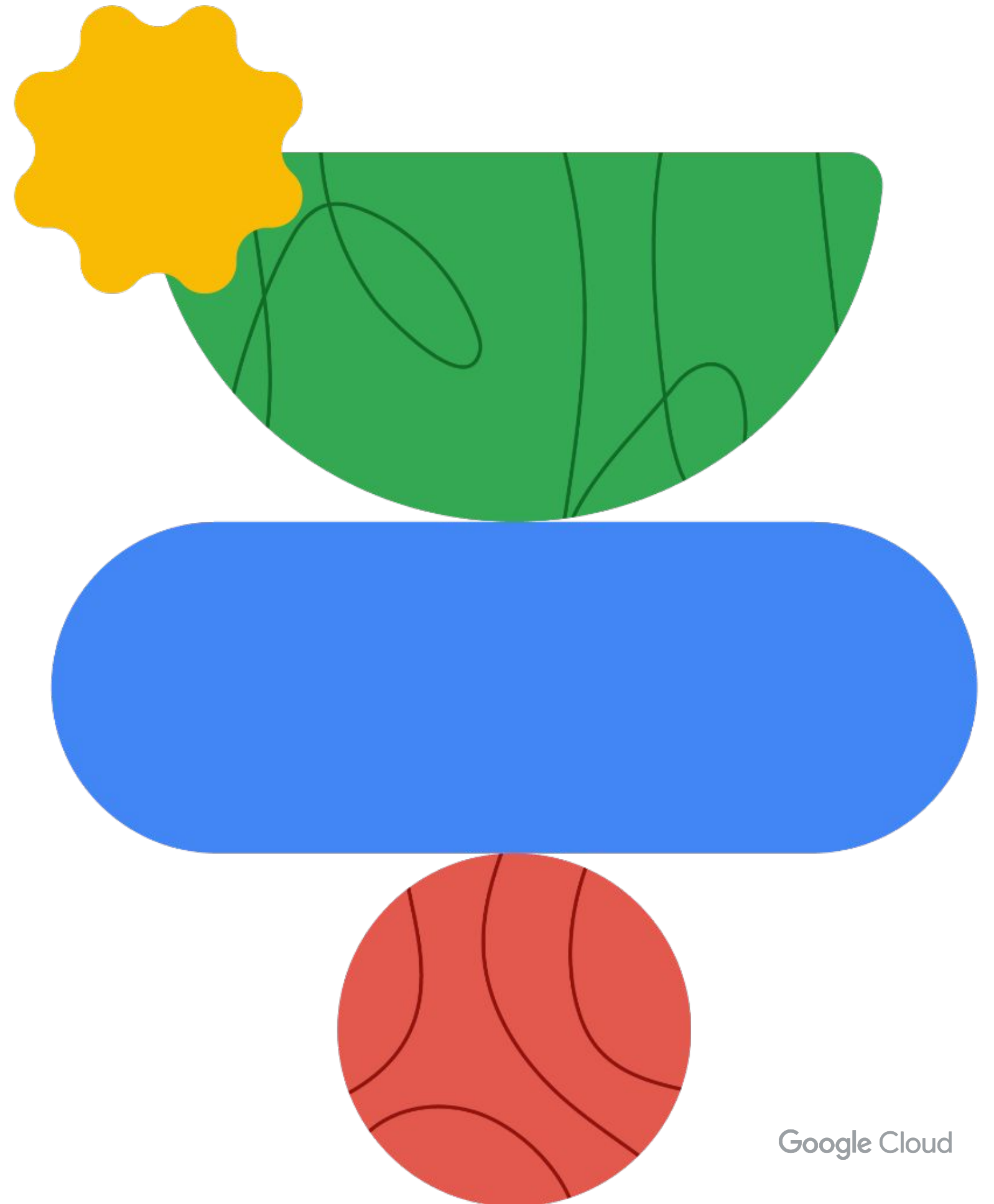
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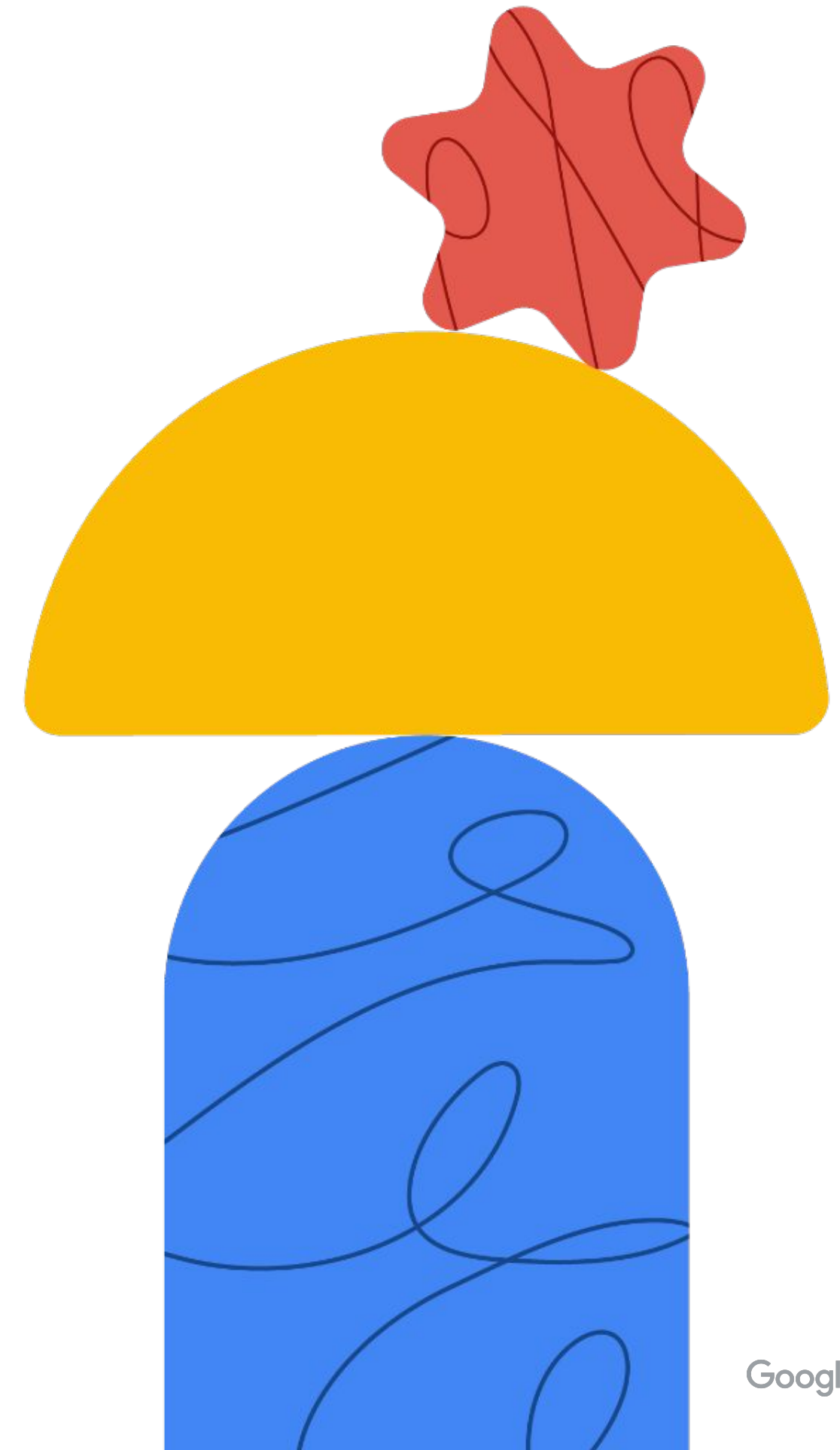
The benefits of running compute workloads in the cloud



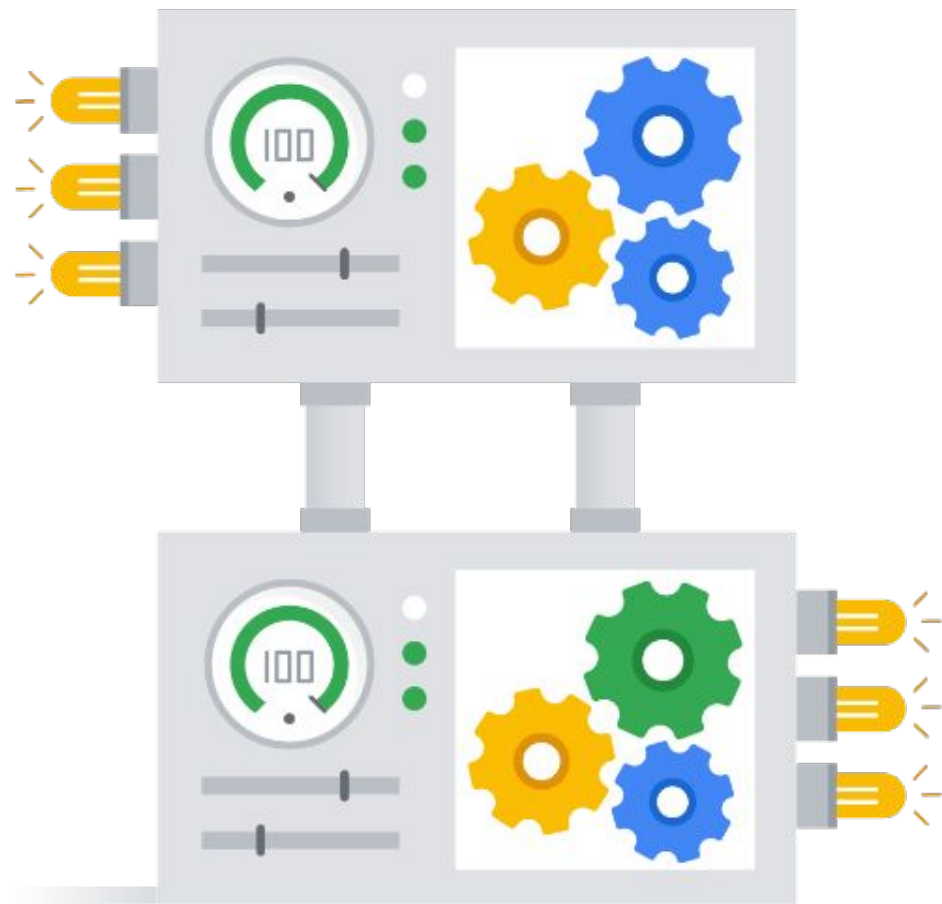
Recall the benefits of running compute workloads in the cloud



Virtual machines



Virtualization is a form of resource optimization that lets multiple systems run on the same hardware

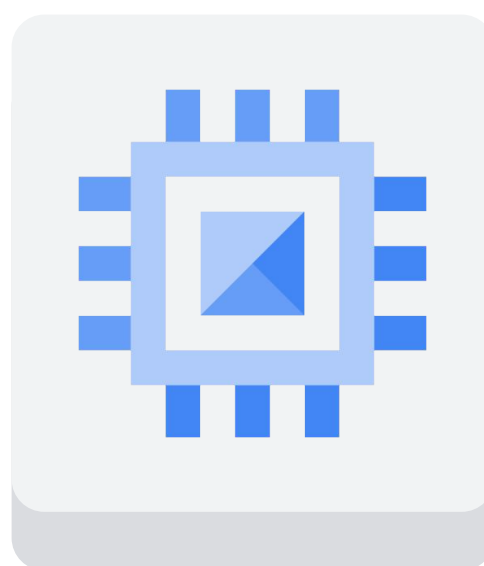


Virtual machines (VMs)

This means that they share the same:

- ✓ Pool of processing
- ✓ Storage
- ✓ Networking resources

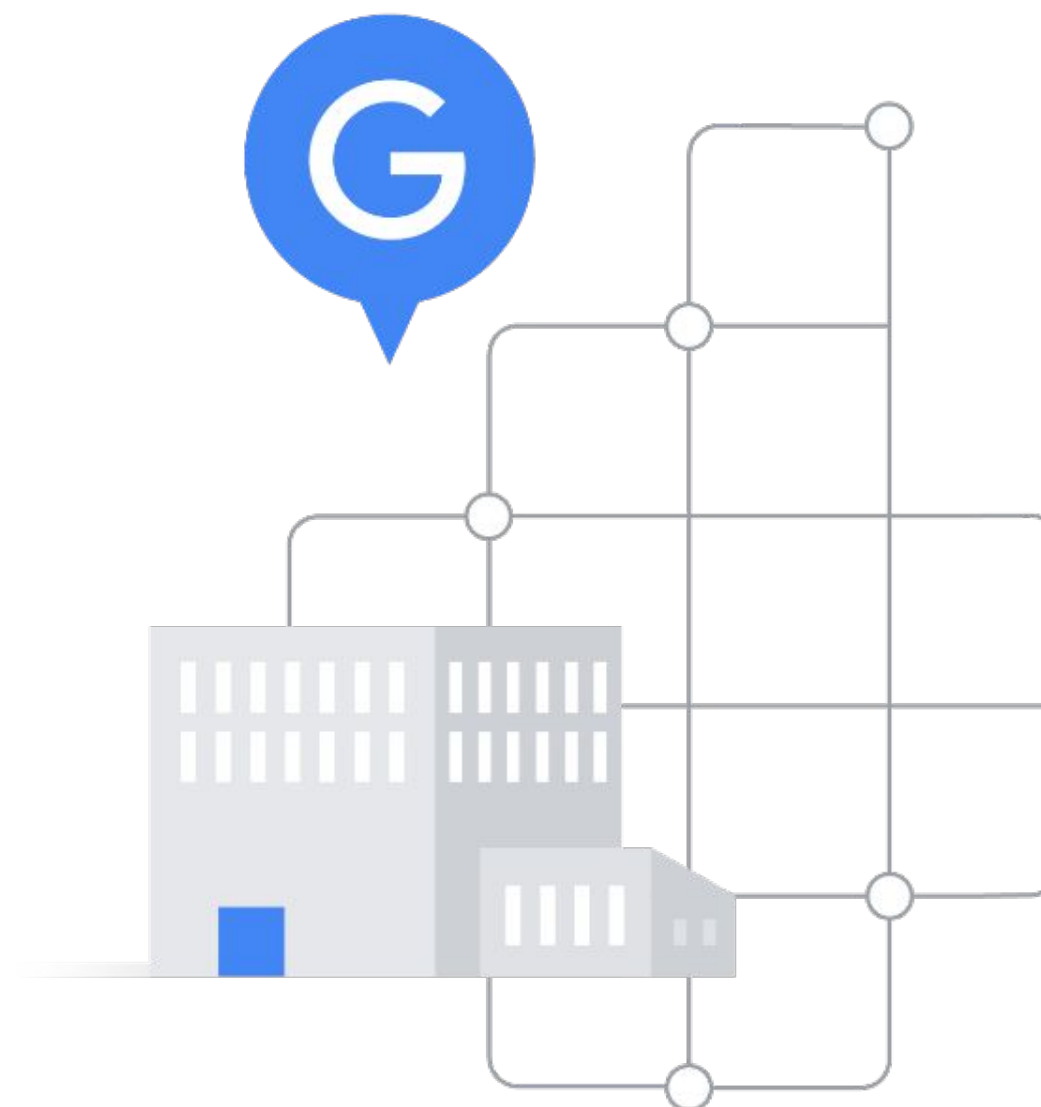
Compute Engine



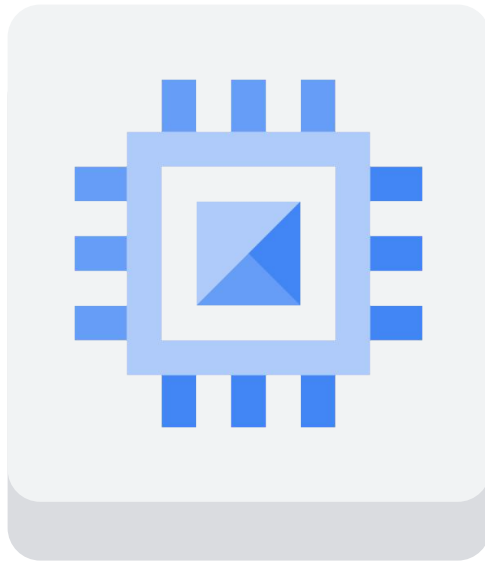
Compute Engine

IaaS solution

Lets users create and run virtual machines on Google infrastructure.



Compute Engine



Compute Engine



There are no upfront investments.



Thousands of virtual CPUs can run on a system that's designed to be fast and to offer consistent performance.



Each VM contains the power and functionality of a full-fledged operating system.



A VM can be configured much like a physical server by specifying:



The amount of CPU power and memory needed

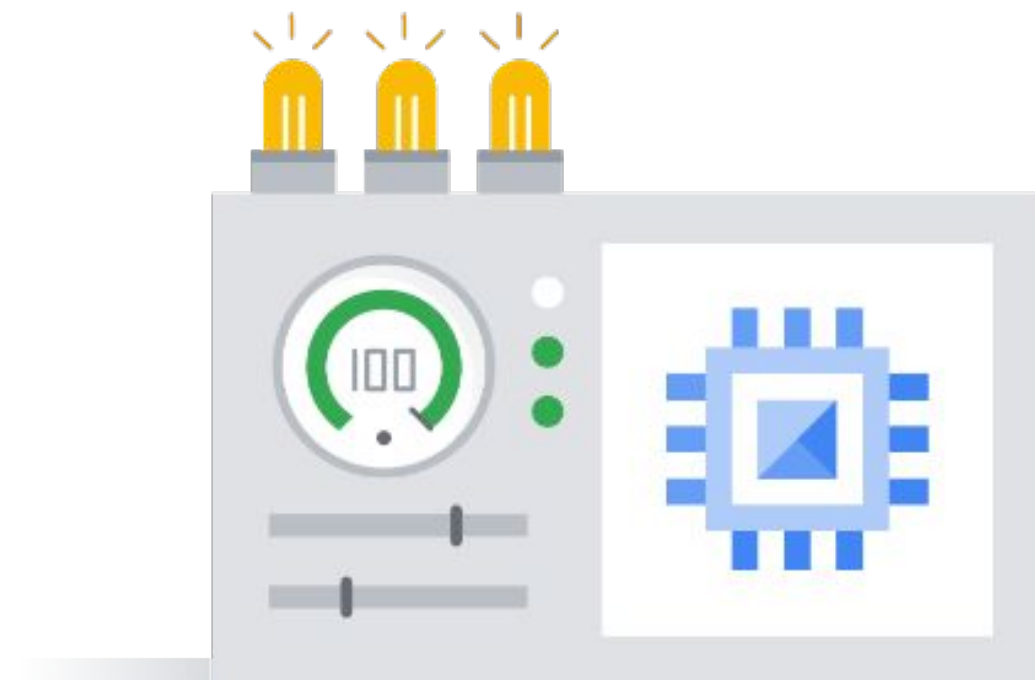


The amount and type of storage needed



The operating system

Virtual machine instances



Can be created through the:

01

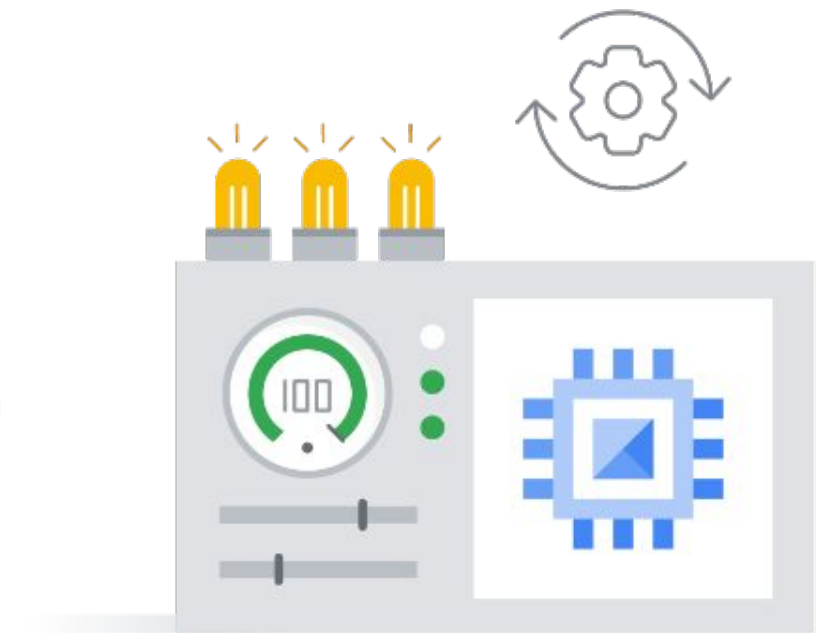
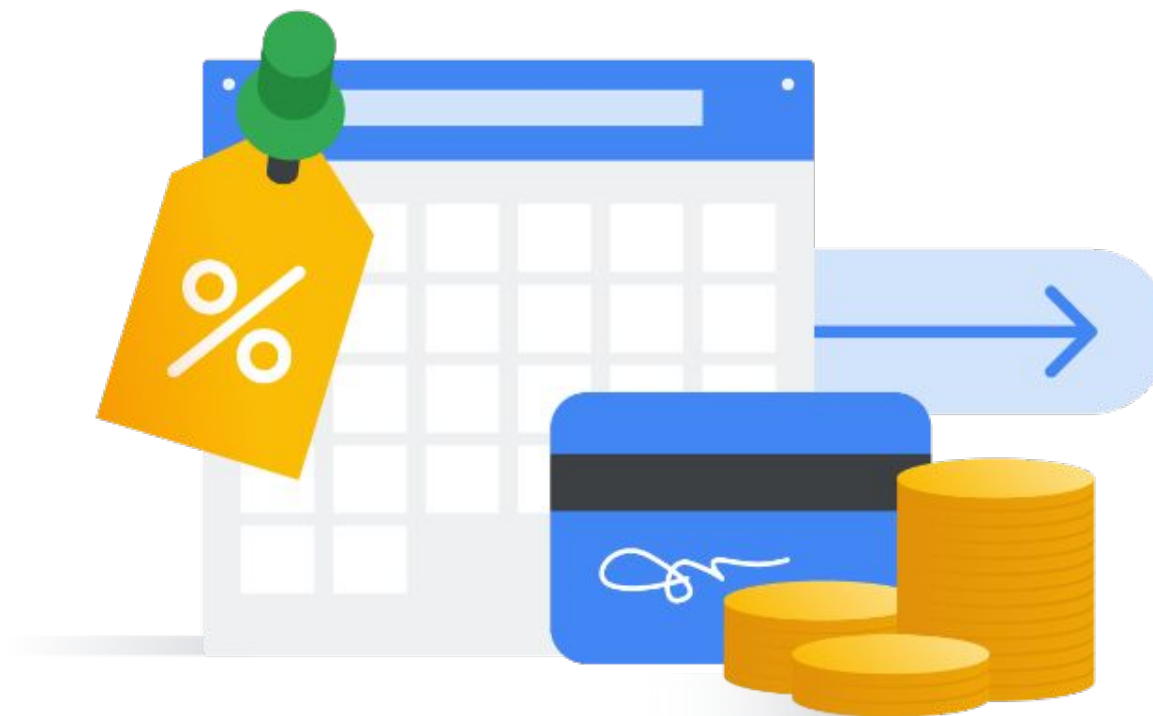
Google Cloud console, which is a web-based tool to manage Google Cloud projects and resources

02

Google Cloud CLI (command-line interface) by using infrastructure automation tools such as Terraform or the Compute Engine API

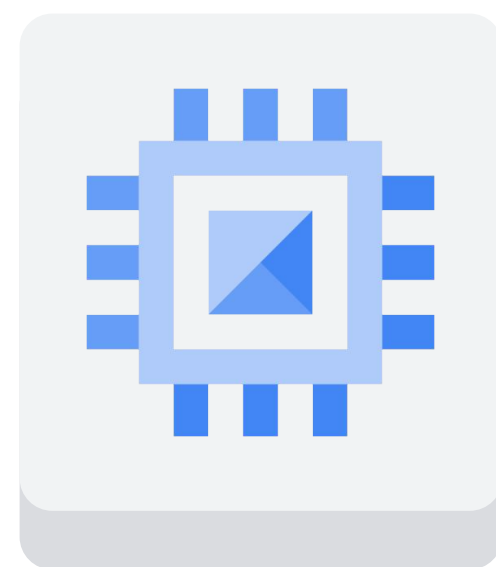
Sustained-use discounts

- ✓ Bills by the second
- ✓ 1 minute minimum
- ✓ Sustained-use discounts
 - They apply automatically to VMs the longer they run.
 - For each VM that runs for more than 25% of a month, Compute Engine applies a discount for every incremental hour of use.

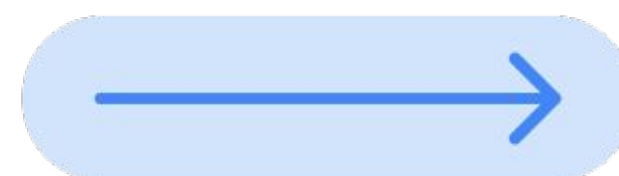


Virtual Machine

Committed-use discounts



Compute Engine



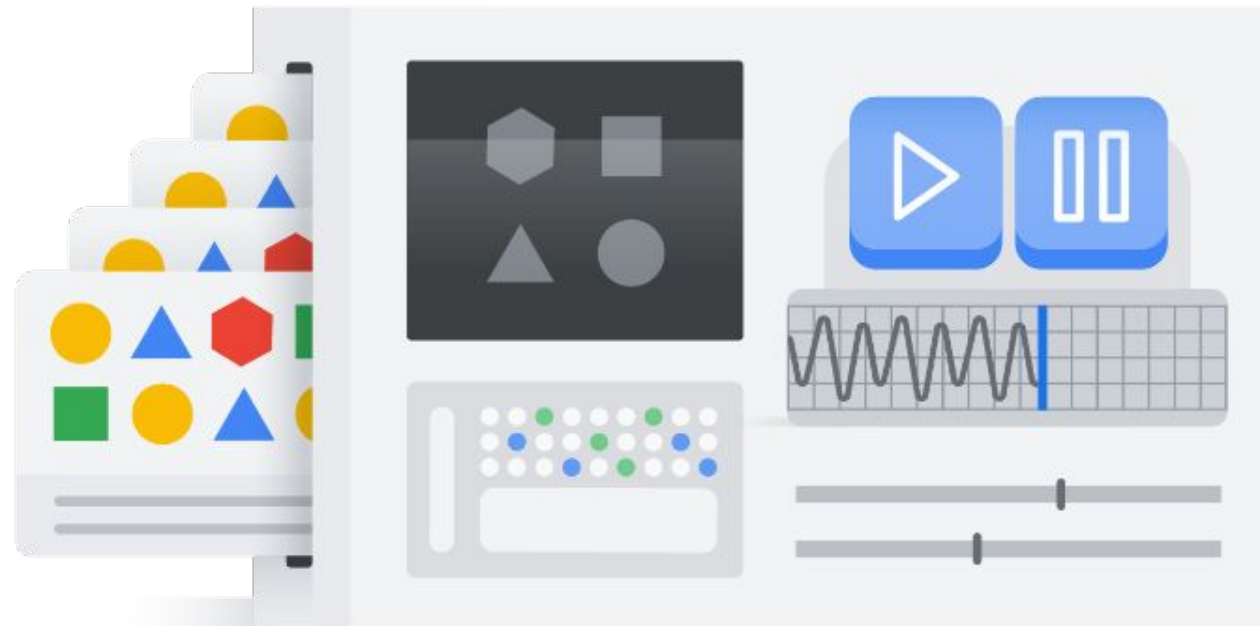
When committing to use resources for either a 1-year or 3-year period



Committed-use discounts

Preemptible and Spot VMs

Example: batch job analyzing a large dataset



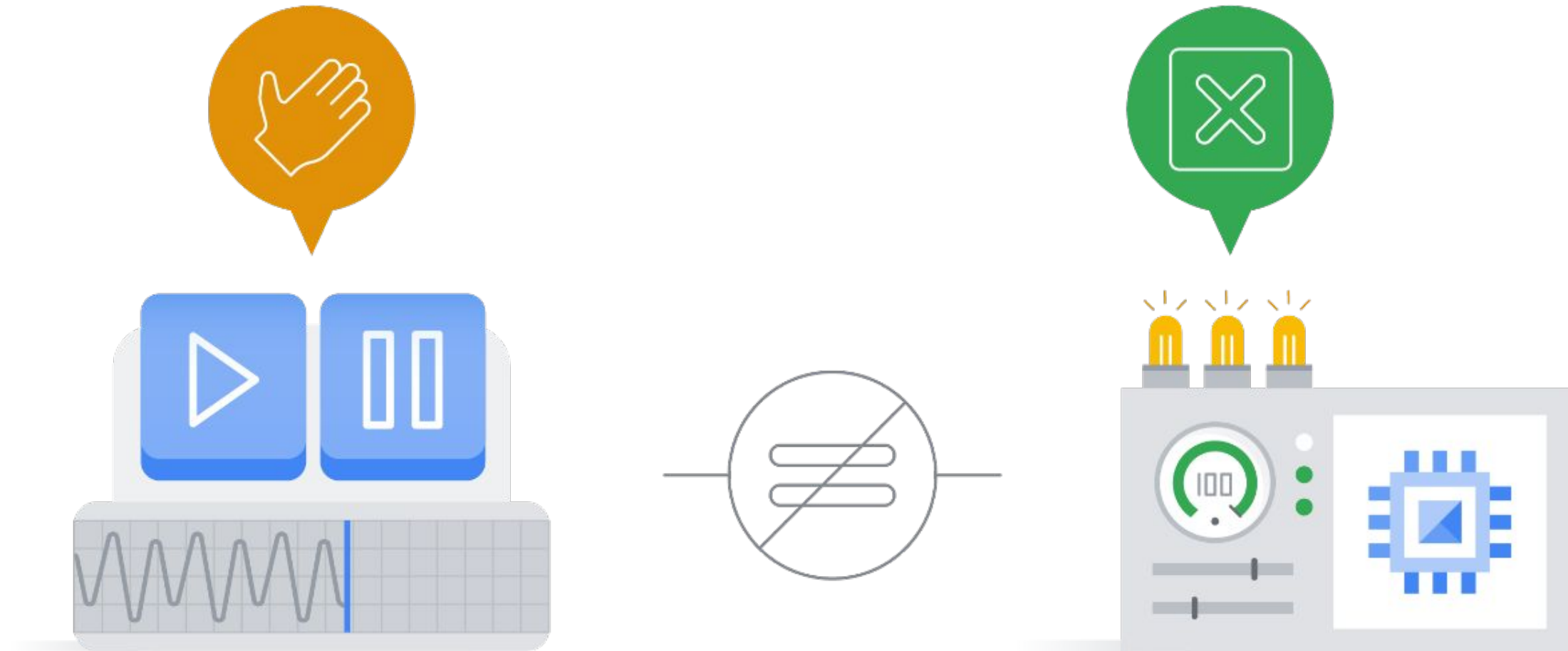
Preemptible or Spot VMs



Costs reduced
up to 90%



Preemptible and Spot VMs: Ensure that a job can be stopped and restarted without impact

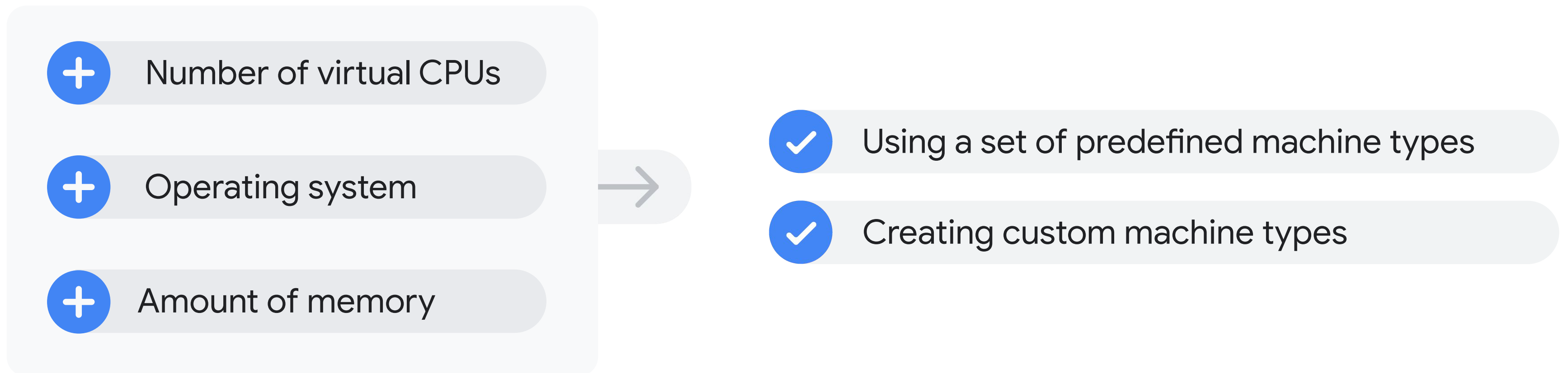


Preemptible or Spot VM

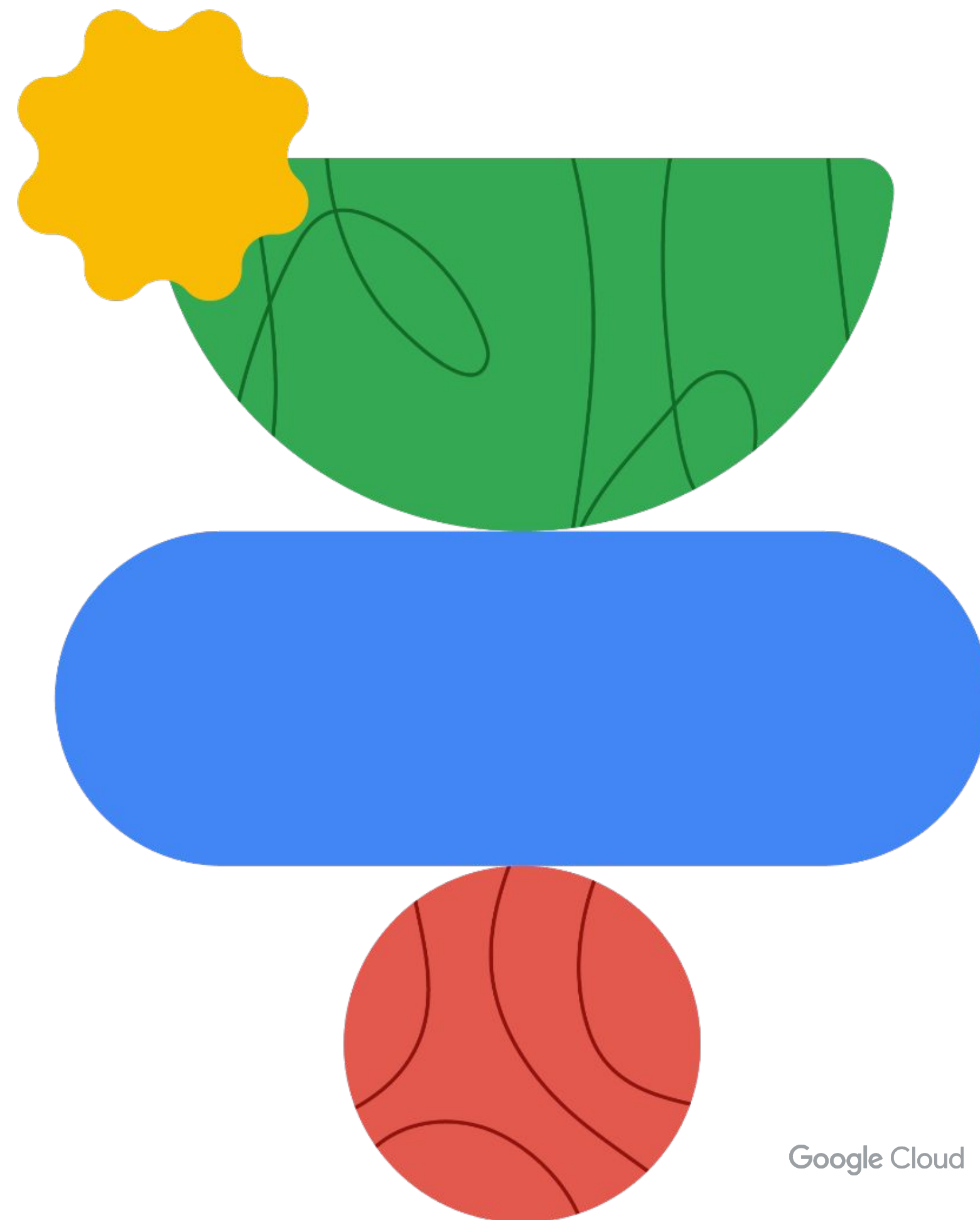
Compute Engine VM

Compute Engine has permission to terminate a VM if its resources are needed elsewhere.

Compute Engine lets users choose the machine properties of their instances



Containers



What is a container?



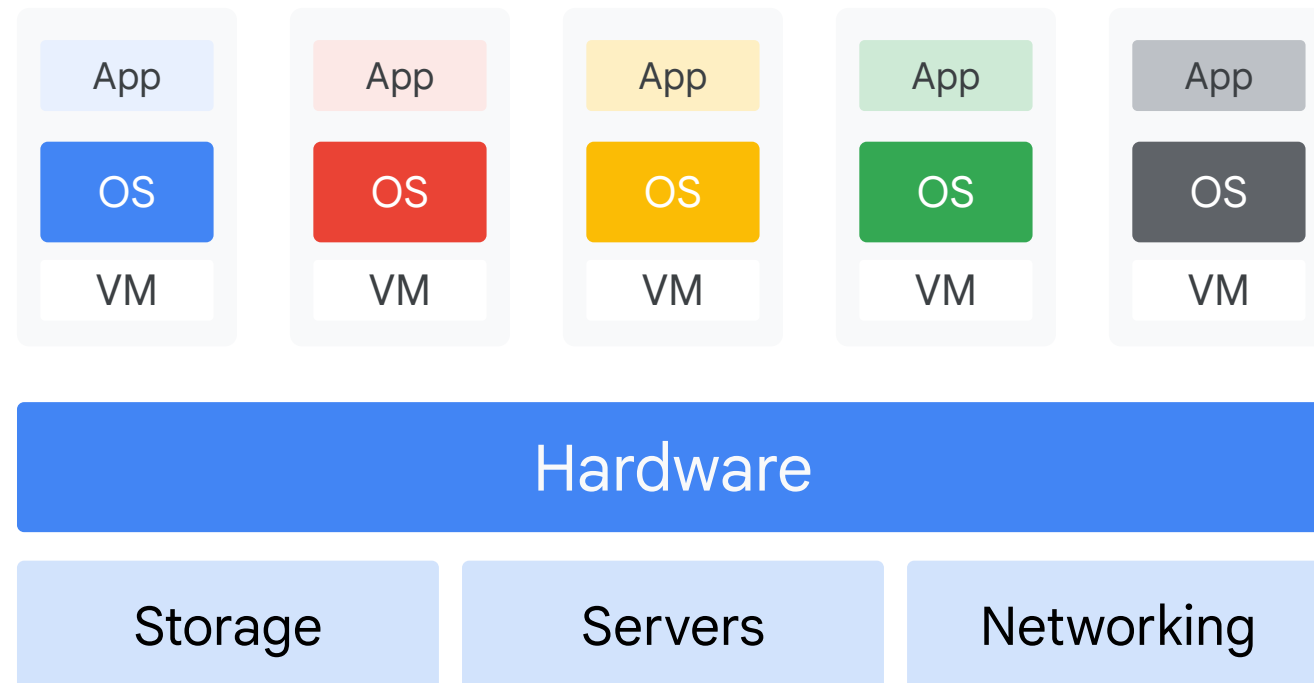
Containers

Containers provide **isolated environments** to run software services and optimize resources from **one piece of hardware**.

The difference between a VM and a container

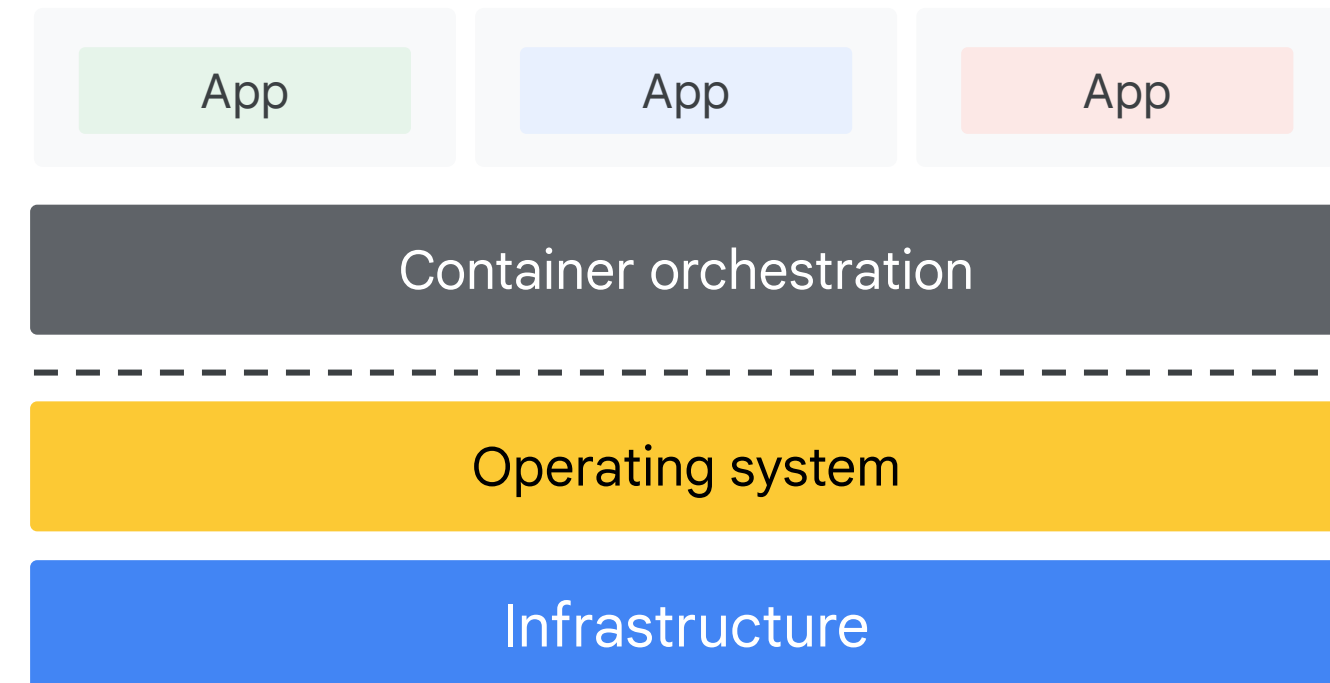
Virtual machines

virtualize an **entire machine** down to the hardware layers



Containers

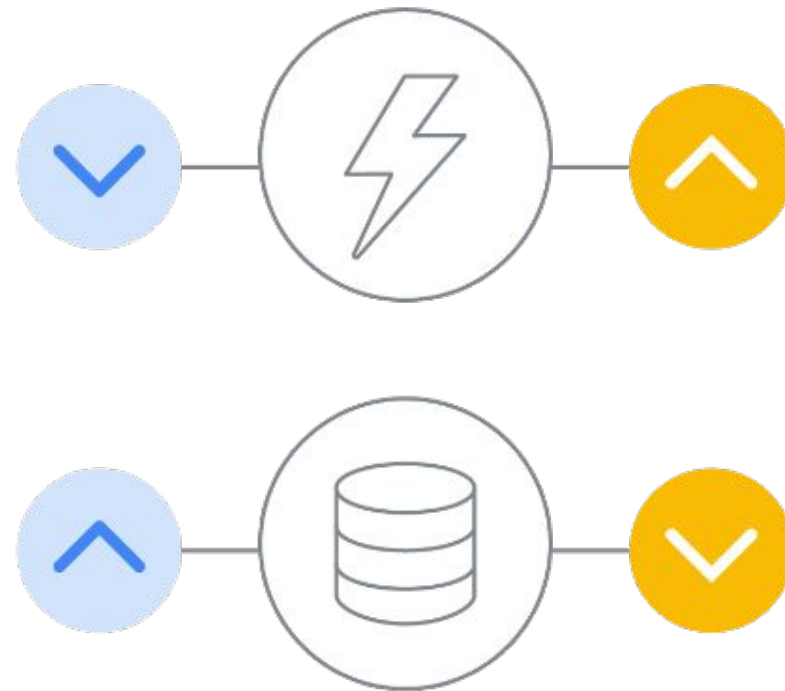
virtualize **software layers** above the operating system level



Containers start faster and use a fraction of the memory compared to booting an entire operating system

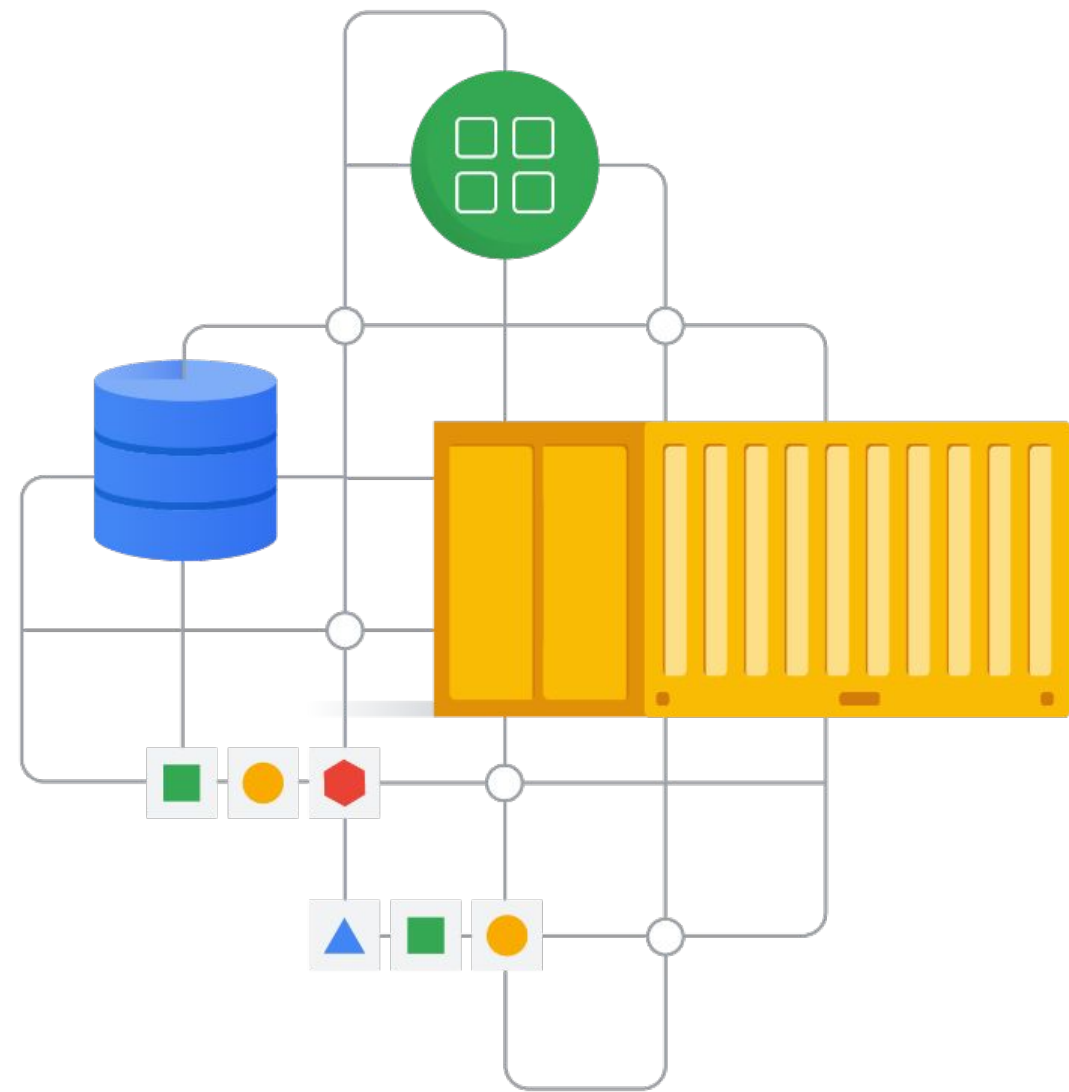


Operating system



Containers

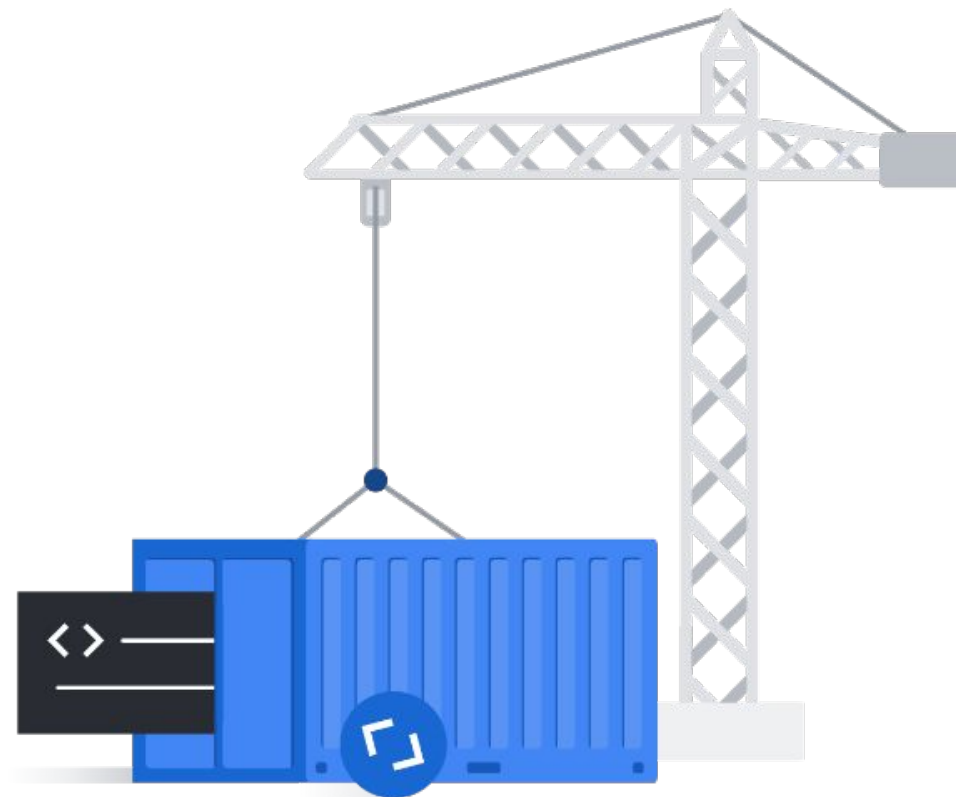
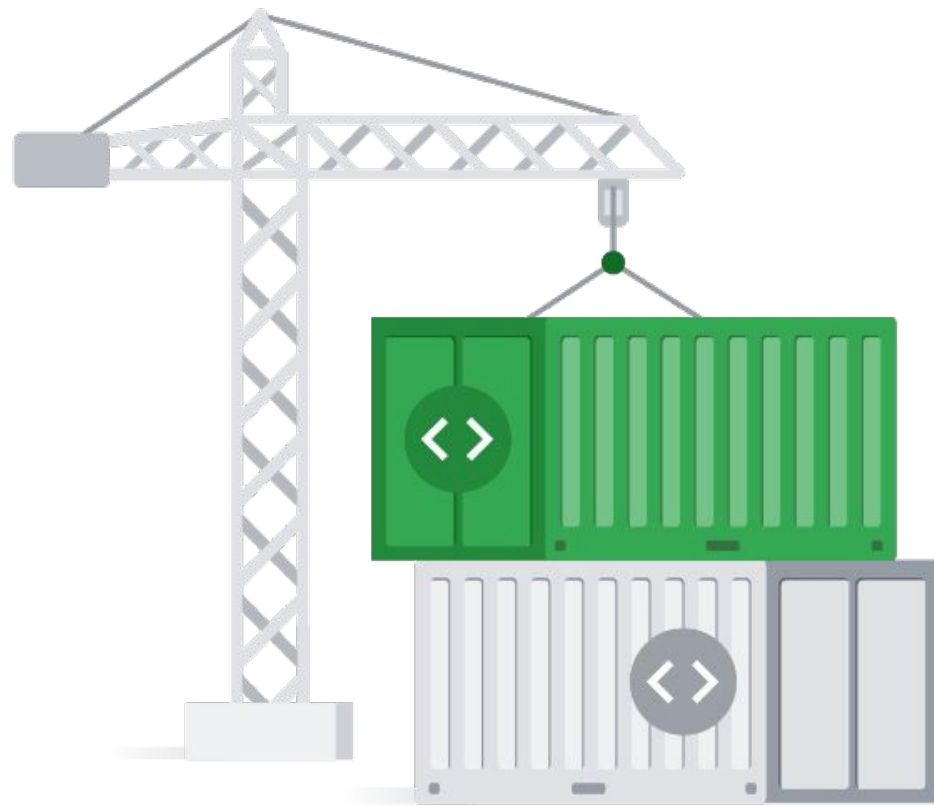
A container is packaged with the application and all of its dependencies



Containers can be:

- ✓ Independently developed
- ✓ Independently tested
- ✓ Independently deployed
- ✓ Well suited for a microservices-based architecture

Containers let developers create predictable environments isolated from other system resources



Developers can easily update a particular part of an application without affecting the rest of it.

Quiz

Question

A travel company is in the early stages of developing a new application and wants to test it on a variety of configurations: different operating systems, processors, and storage options. What cloud computing option should they use?

- A. Containers
- B. A local development environment
- C. Colocation
- D. Virtual machine instances

Quiz

Answer

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Quiz

Question

What portion of a machine does a container virtualize?

- A. Software layers above the firmware level.
- B. Software layers above the operating system level.
- C. Hardware layers above the electrical level.
- D. The entire machine.

Quiz

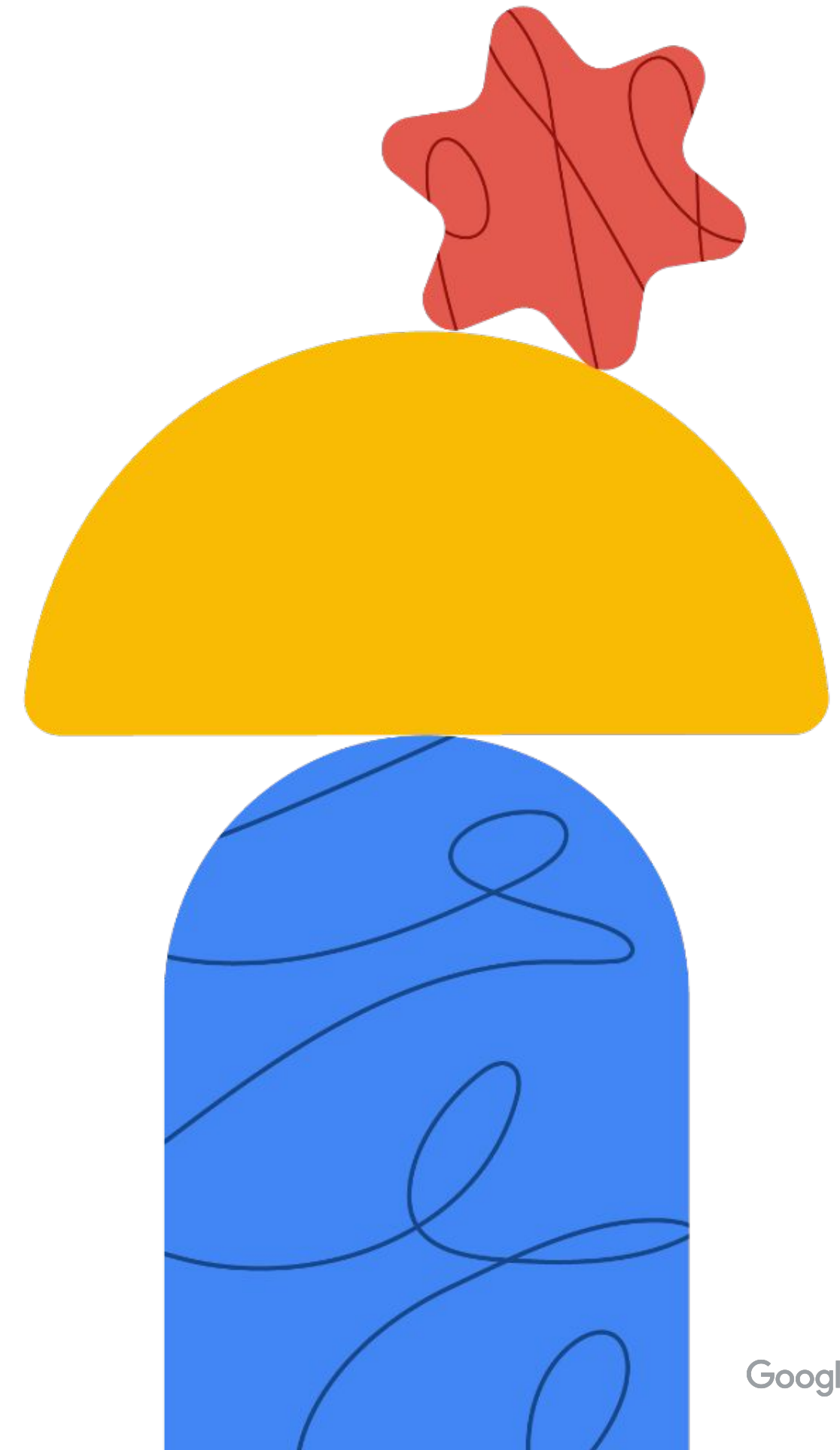
Answer

What portion of a machine does a container virtualize?

- A. Software layers above the firmware level.
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Container management



Organizations need to manage their IT infrastructure as it becomes more complex

Keep containers secure.

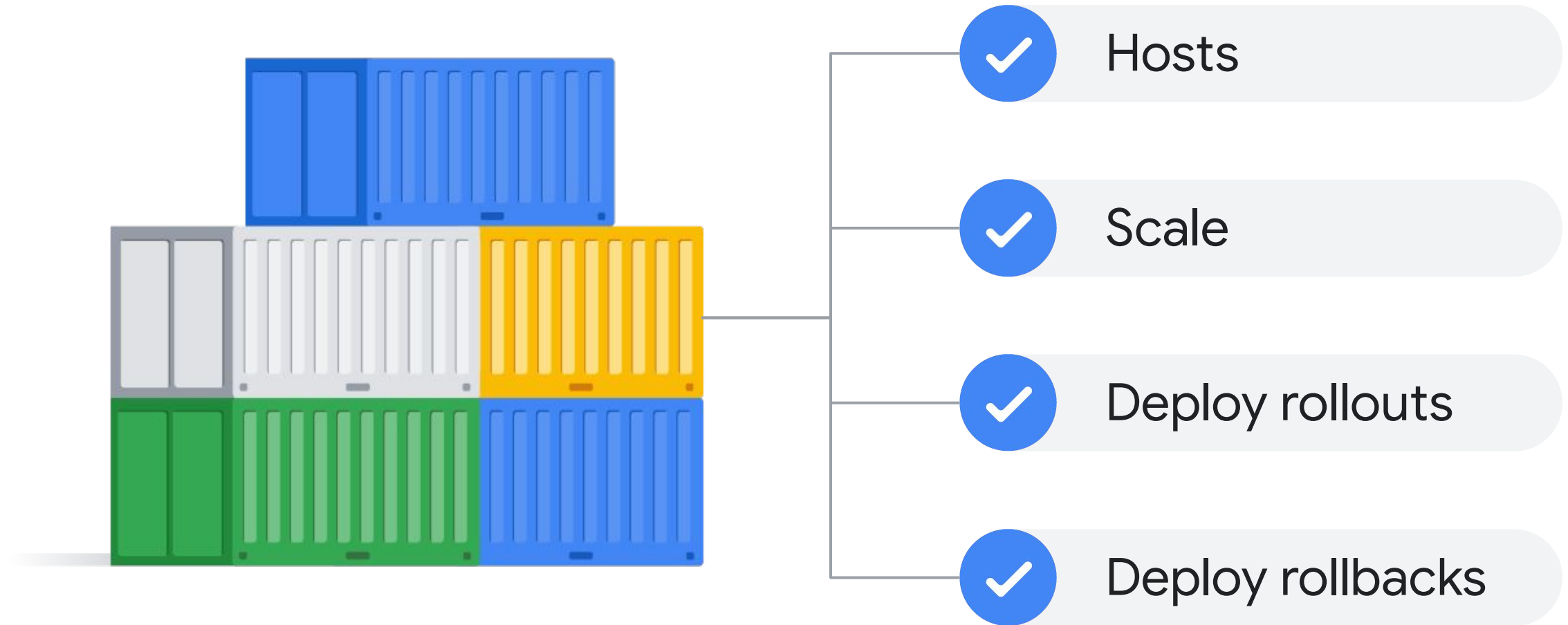
Ensure that containers operate efficiently.



Kubernetes



Is an open-source platform for managing containerized workloads and services.

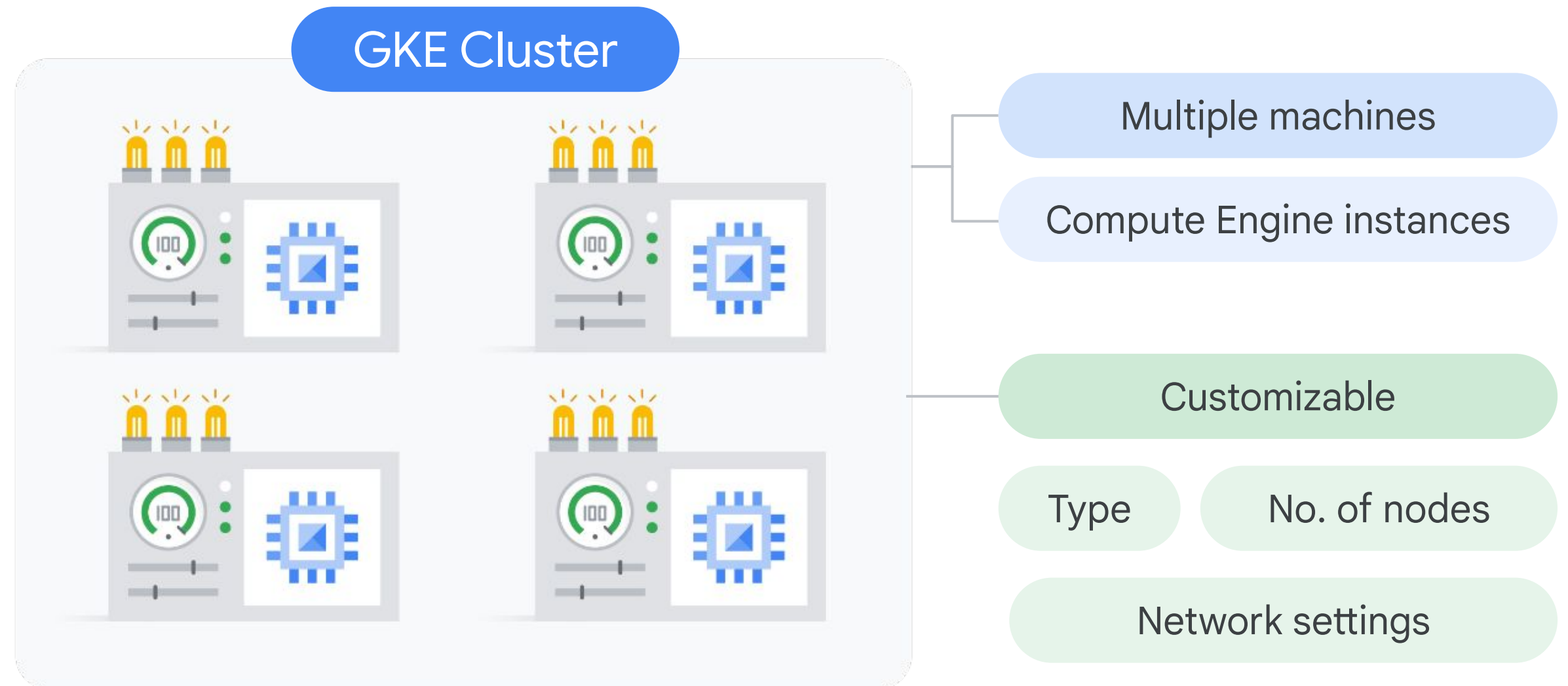


Makes it easy to orchestrate many containers.

Google Kubernetes Engine (GKE)



A Google-hosted
managed
Kubernetes service
in the cloud



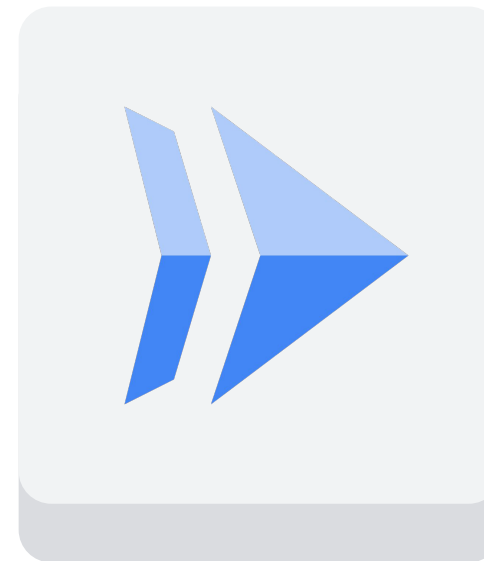
Cloud Run: Another option to run containerized apps on Google Cloud



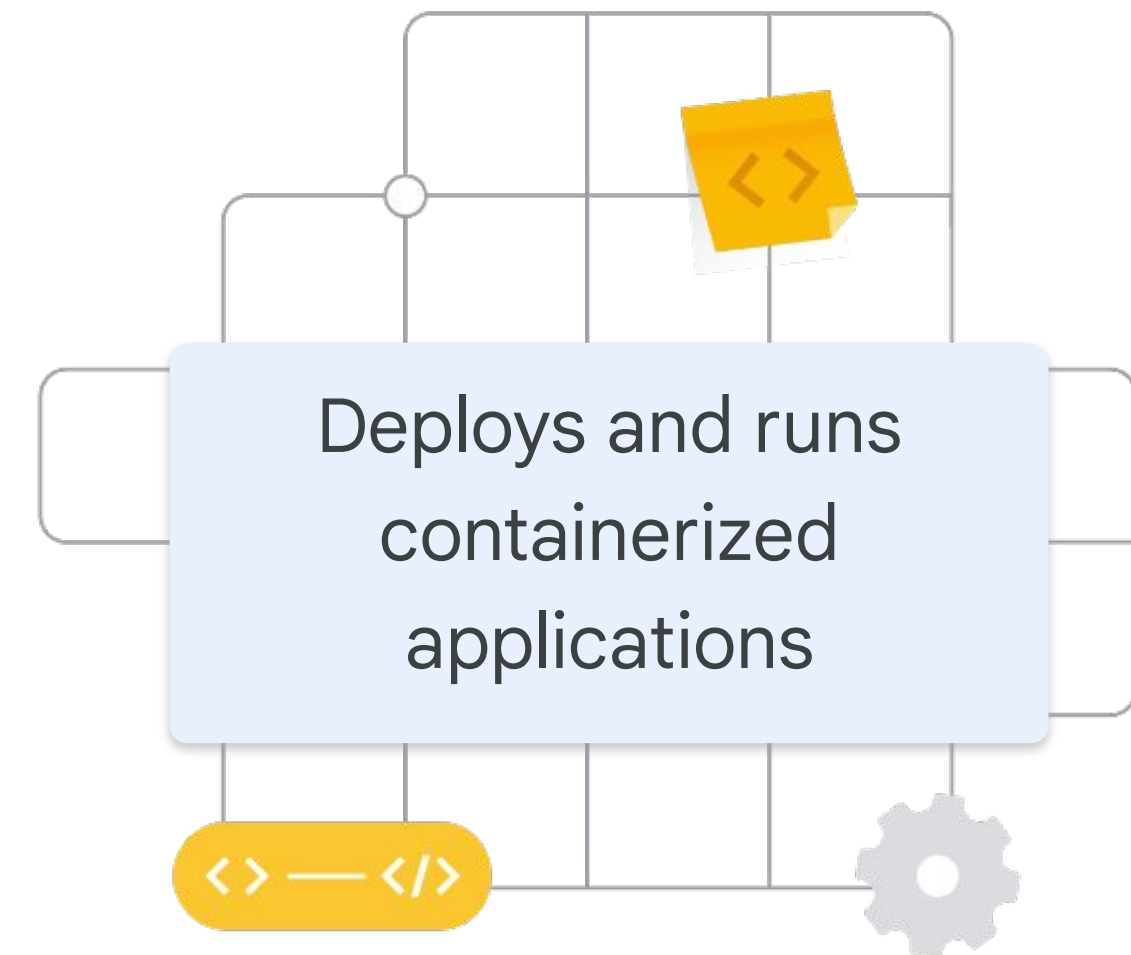
Fully managed



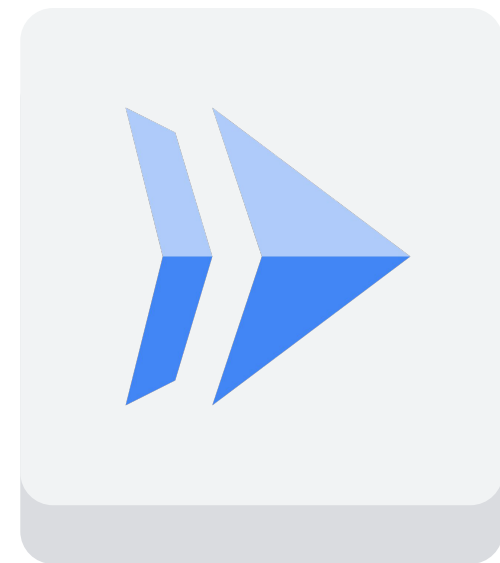
Serverless



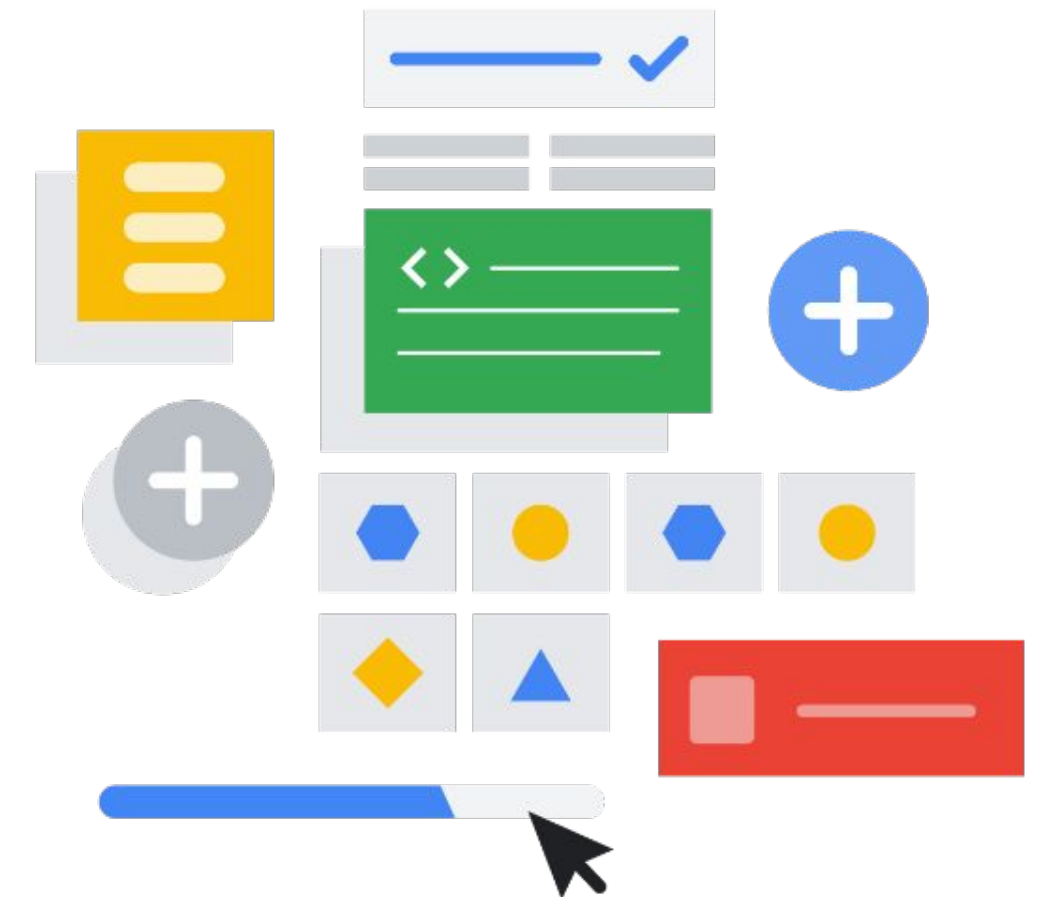
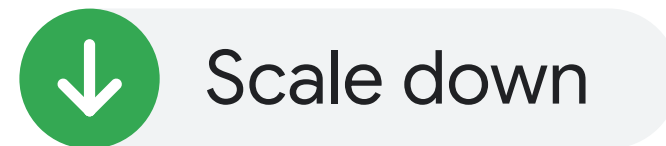
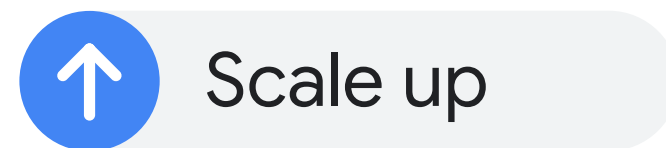
Cloud Run



Cloud Run is ideal for running stateless applications



Cloud Run



Suitable for simple and lightweight applications, such as web applications

Google Kubernetes vs. Cloud Run



GKE

Provides lots of control over a Kubernetes environment with complex applications to run.



Cloud Run

Is a simple, fully managed serverless platform that can scale up and down quickly.

Scenario

Question

An ecommerce company is developing a new web application that will be used by customers to browse products, add items to their carts, and checkout. The company wants to deploy the application that is scalable and reliable, and that can handle a large number of concurrent users during peak periods.

Should the company use **GKE** or **Cloud Run** to deploy the application?

Scenario

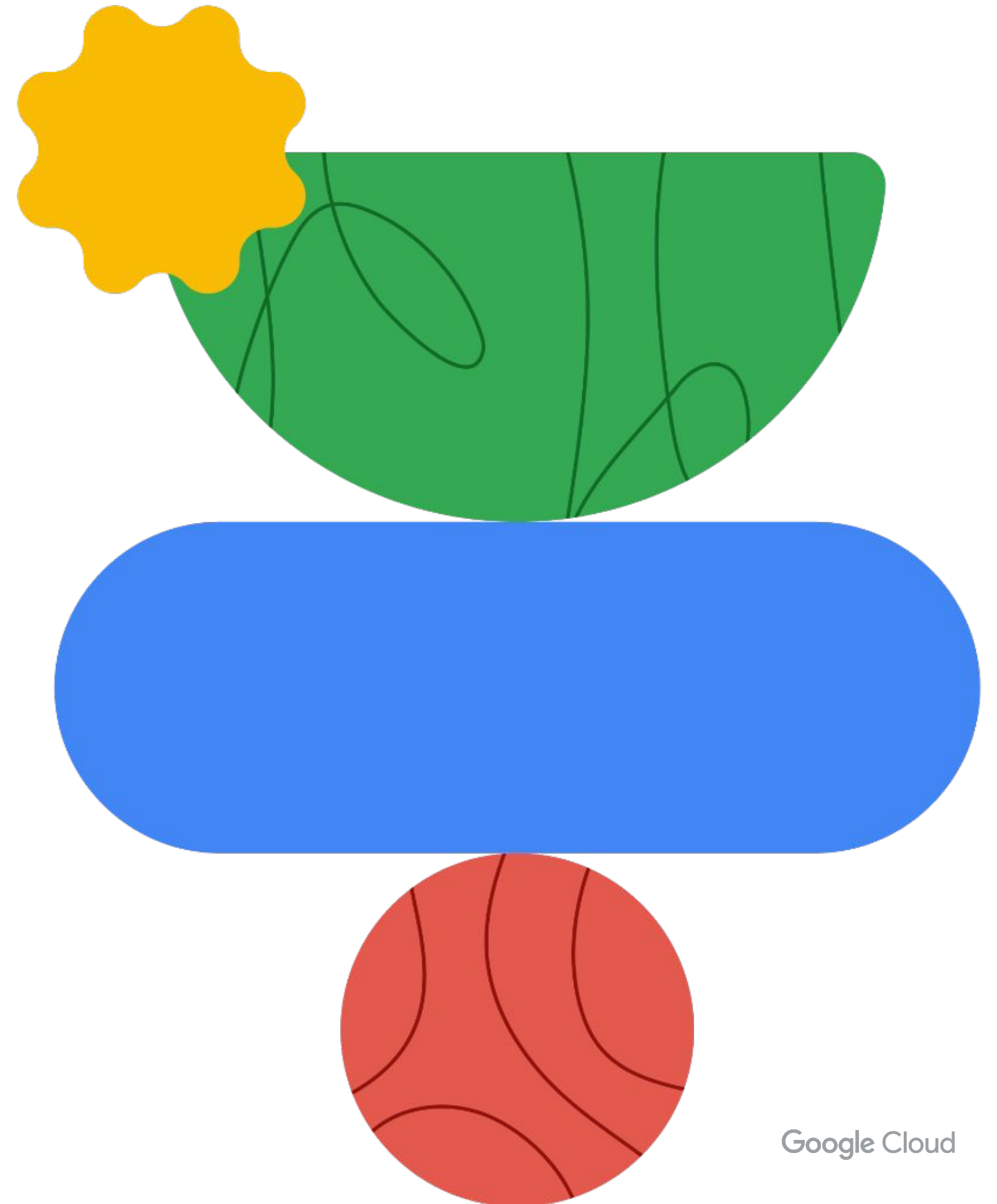
Answer

An ecommerce company is developing a new web application that will be used by customers to browse products, add items to their carts, and checkout. The company wants to deploy the application that is scalable and reliable, and that can handle a large number of concurrent users during peak periods.

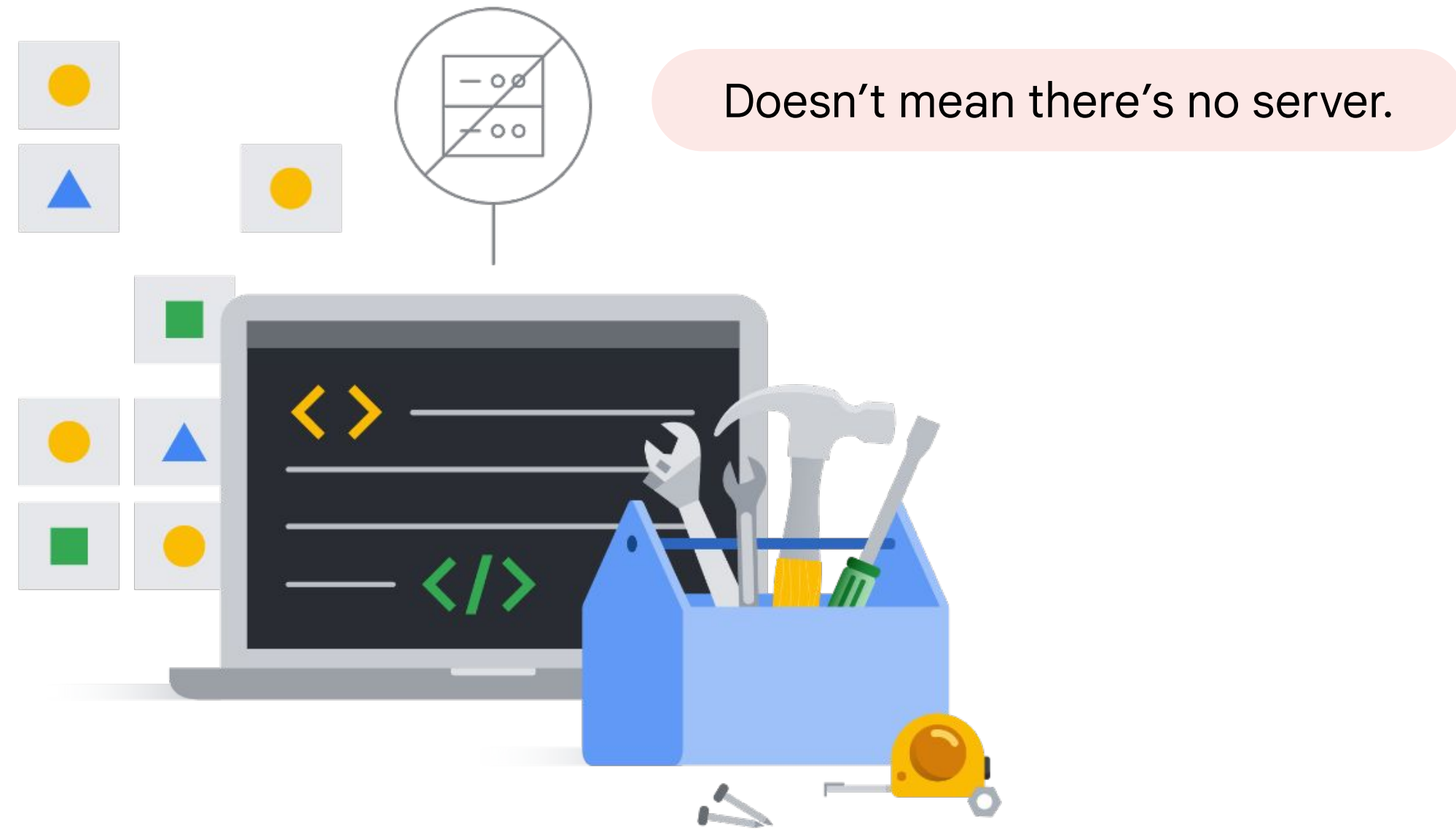
Should the company use **GKE** or **Cloud Run** to deploy the application?

In this scenario, the e-commerce company should choose **Cloud Run**. Cloud Run is ideal for running applications that need to scale up and down quickly in response to traffic. This makes Cloud Run most suitable for simple and lightweight applications, such as web applications. It is also a good choice for companies that do not want to worry about managing infrastructure.

Serverless computing

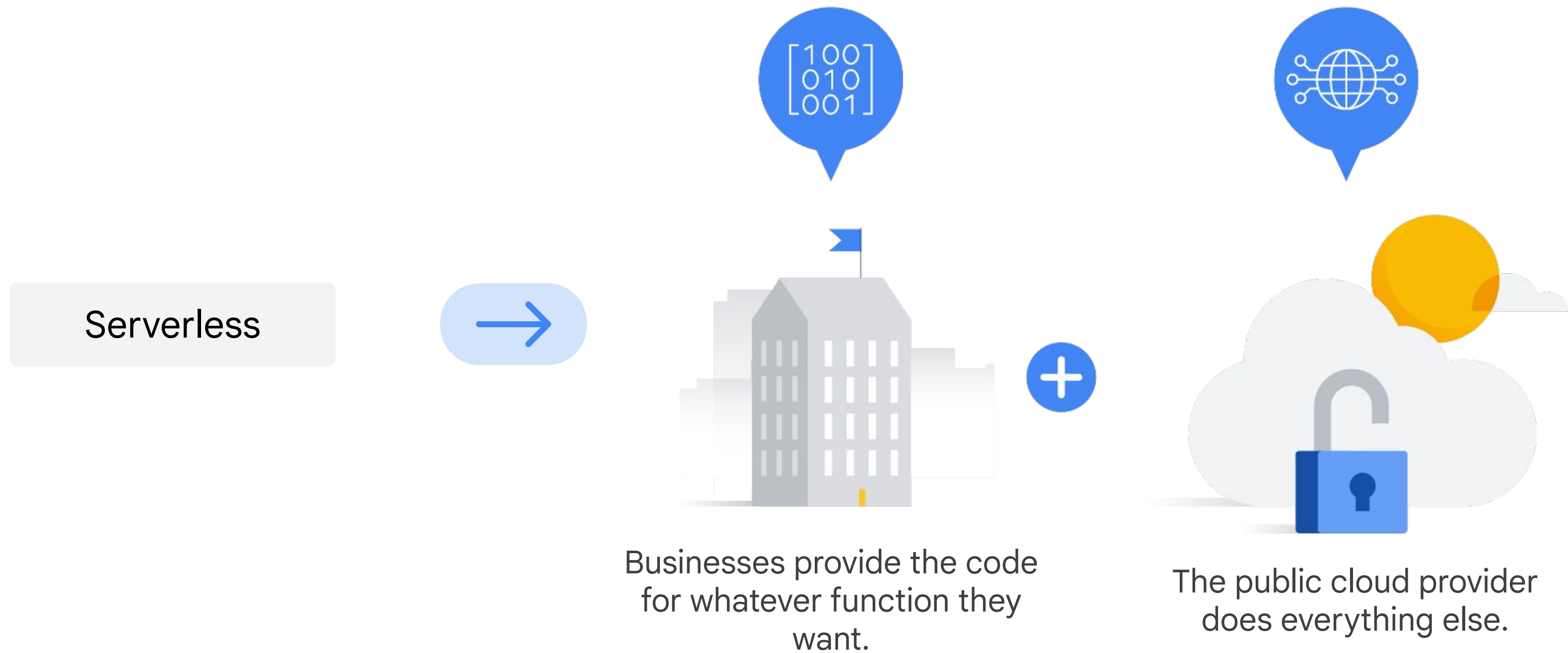


Serverless computing

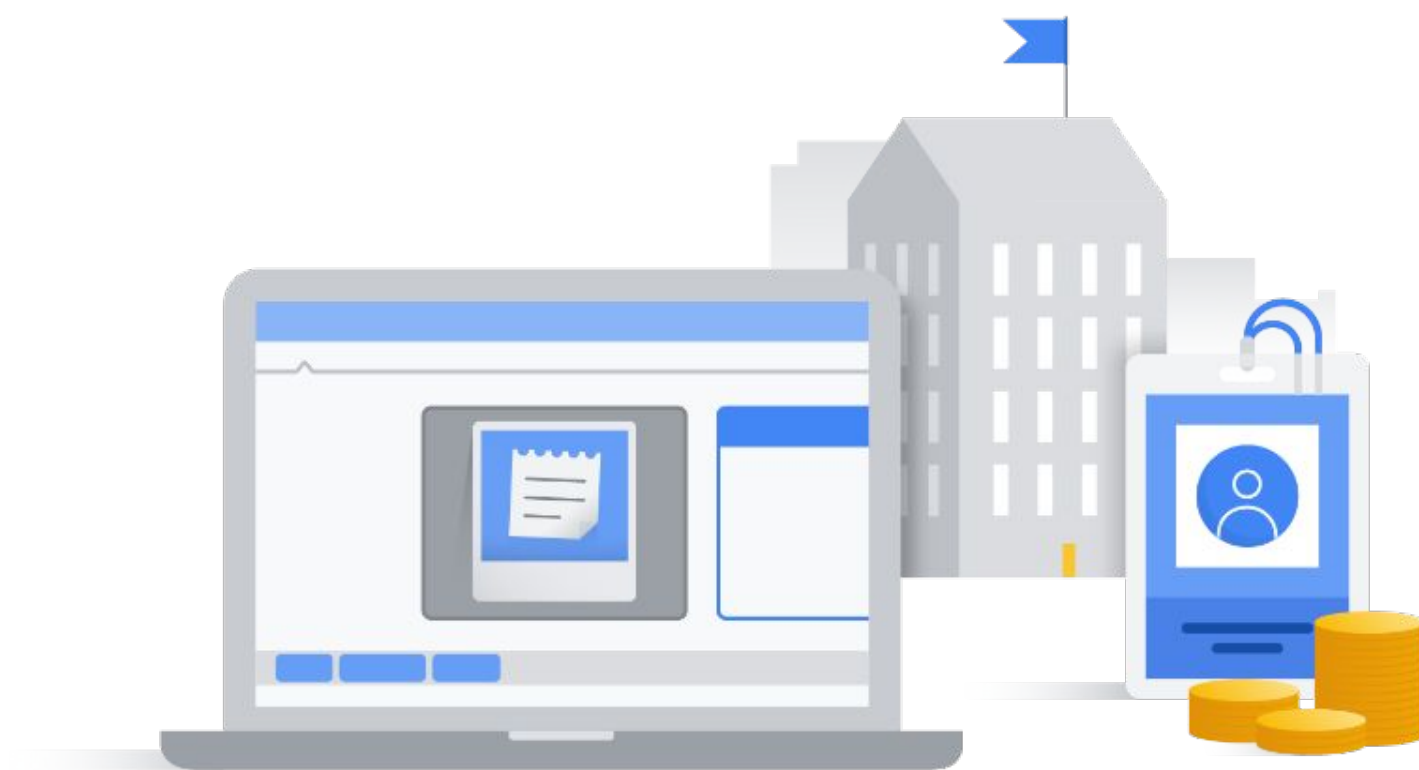


When resources, like compute power, are automatically provisioned in the background as needed.

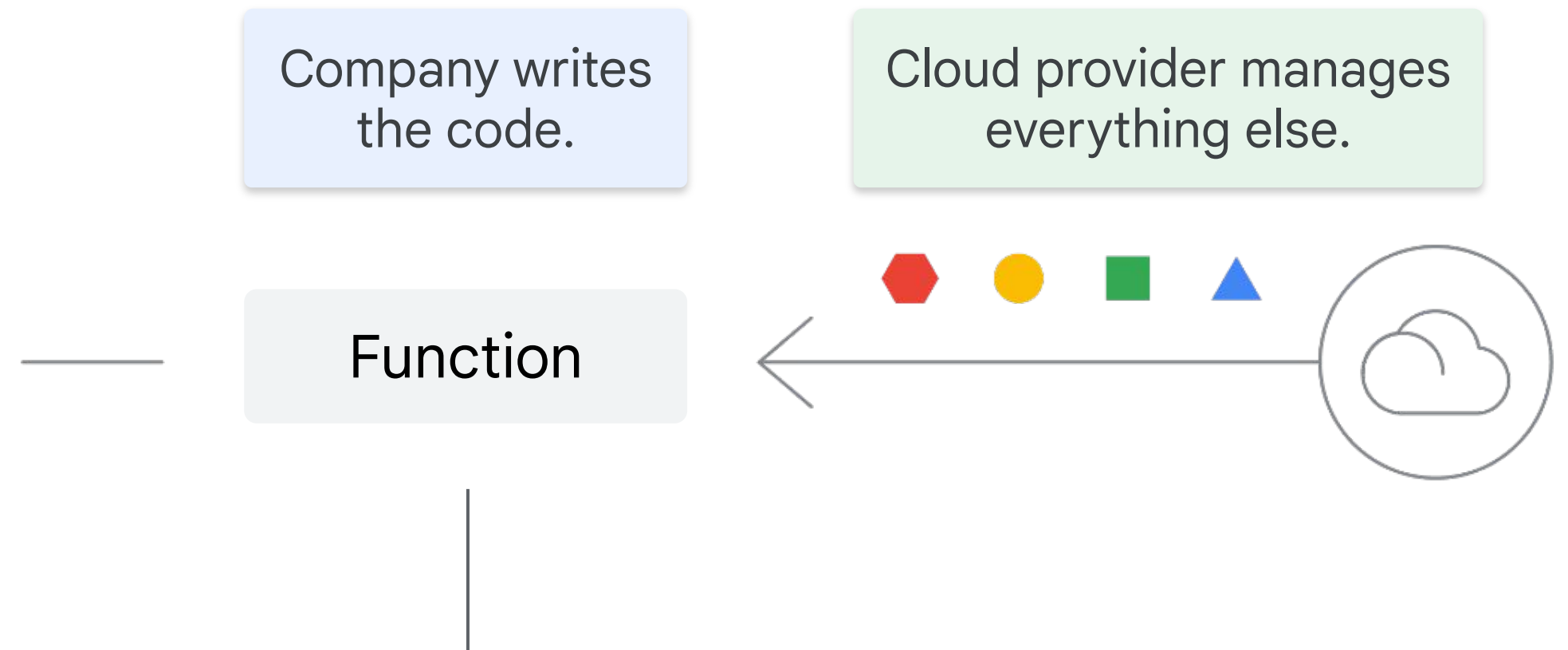
Serverless computing



An example

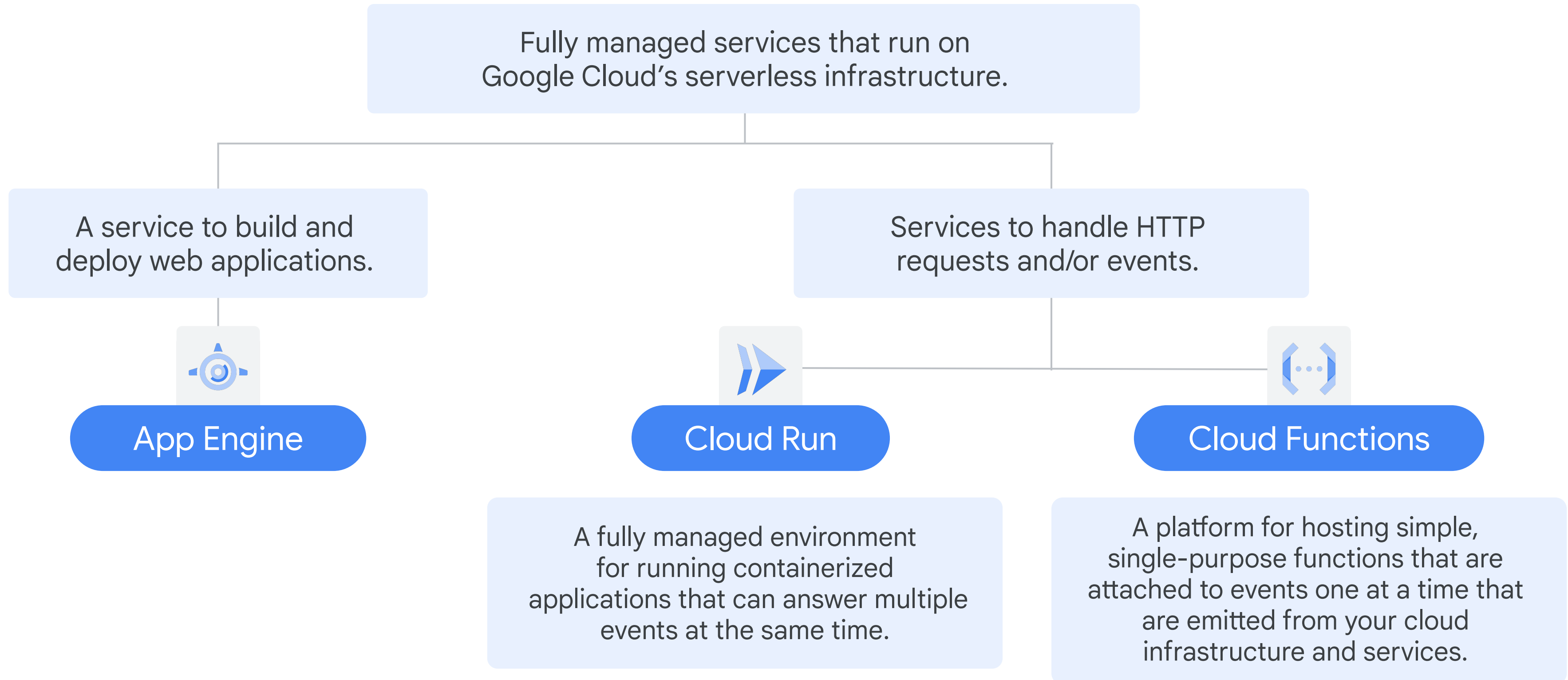


Imagine you provide software to businesses that helps employees manage their corporate expenses.

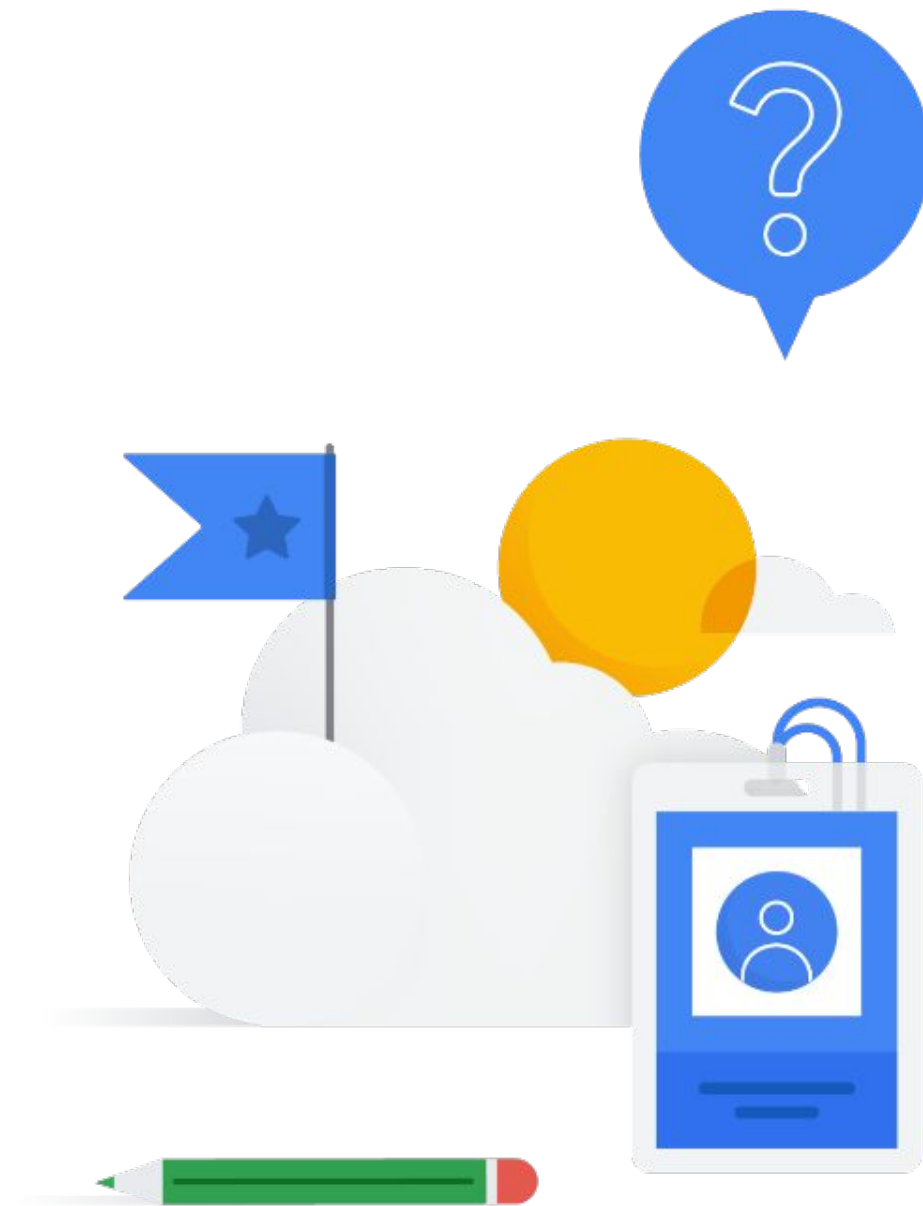


You want to add a feature that lets users upload an image with their expense receipt.

Google Cloud's serverless computing products



Serverless computing benefits



- ✓ Reduced operational costs
- ✓ Scalability
- ✓ Faster time-to-market
- ✓ Reduced development costs
- ✓ Improved resilience
- ✓ Pay-per-use pricing model

Module 4

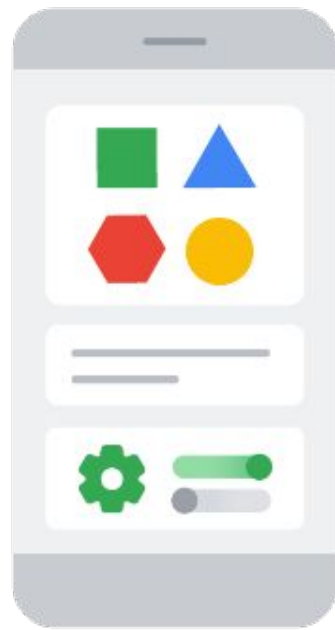
Modernize Infrastructure and Applications with Google Cloud



Lessons

- 01 Important cloud migration terms
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What exactly is an application?

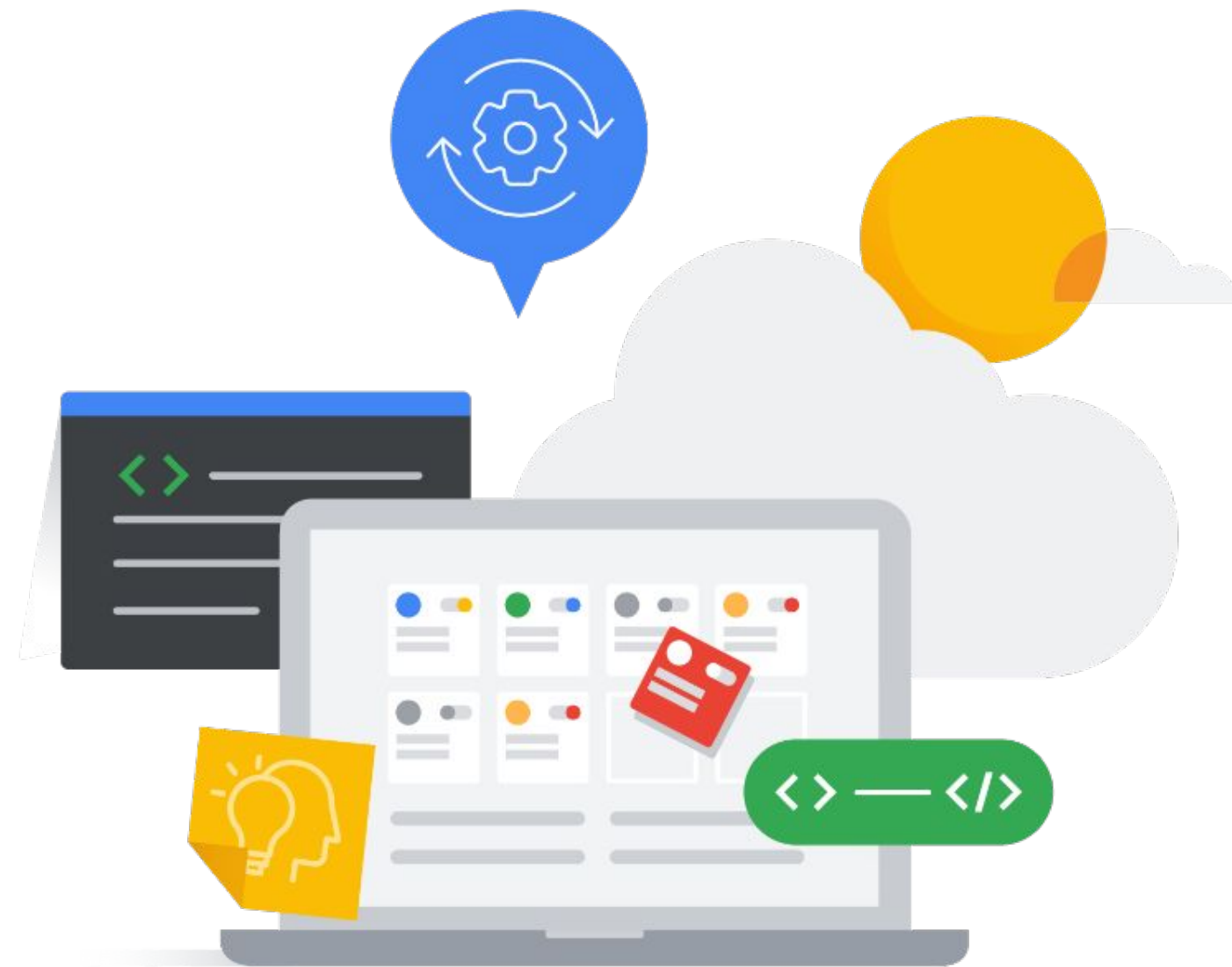


Application



A computer program or software that helps users do something.

Modern cloud application development



Flexible



Scalable



Uses the latest
technologies

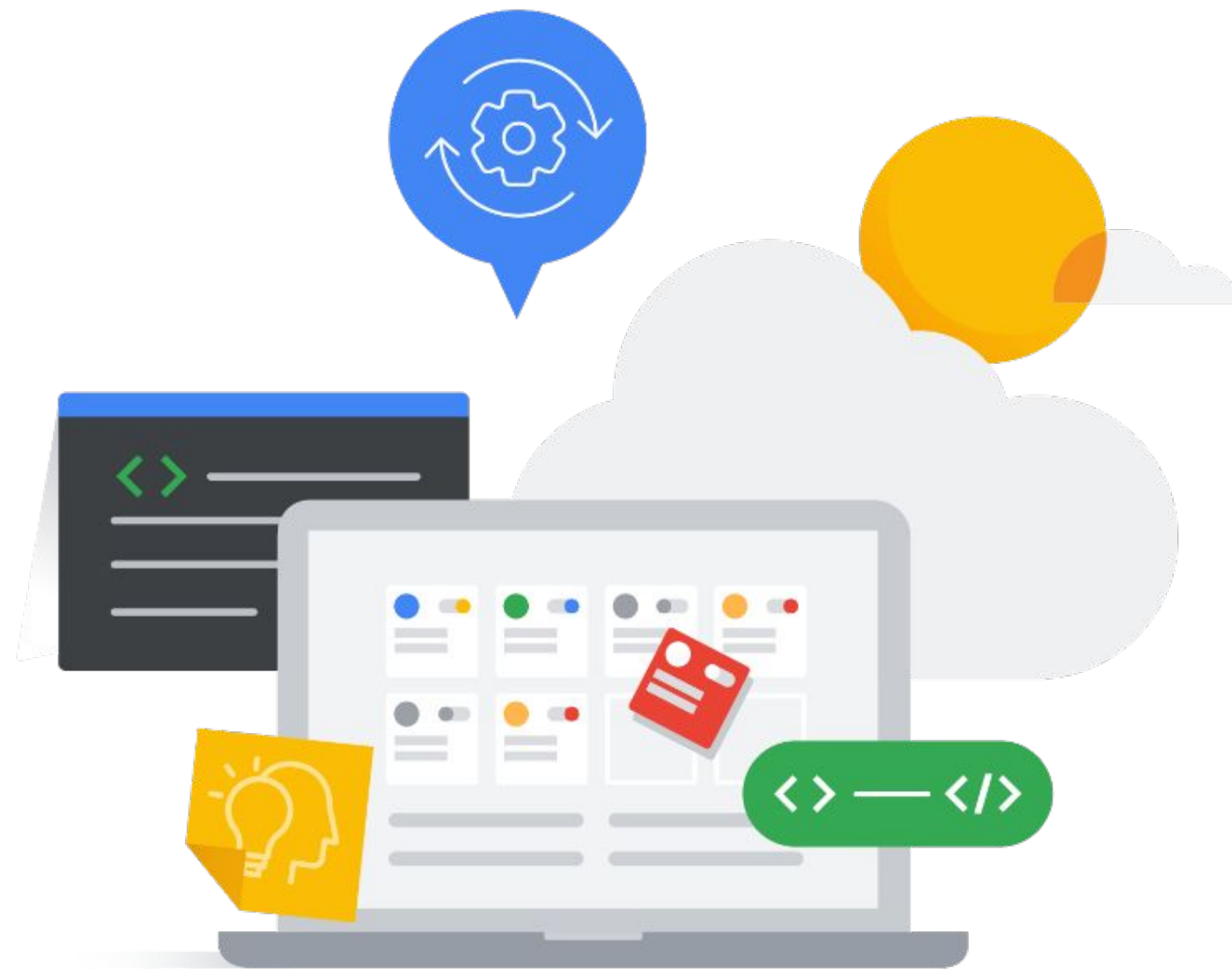
Traditional software development



Monolithic applications

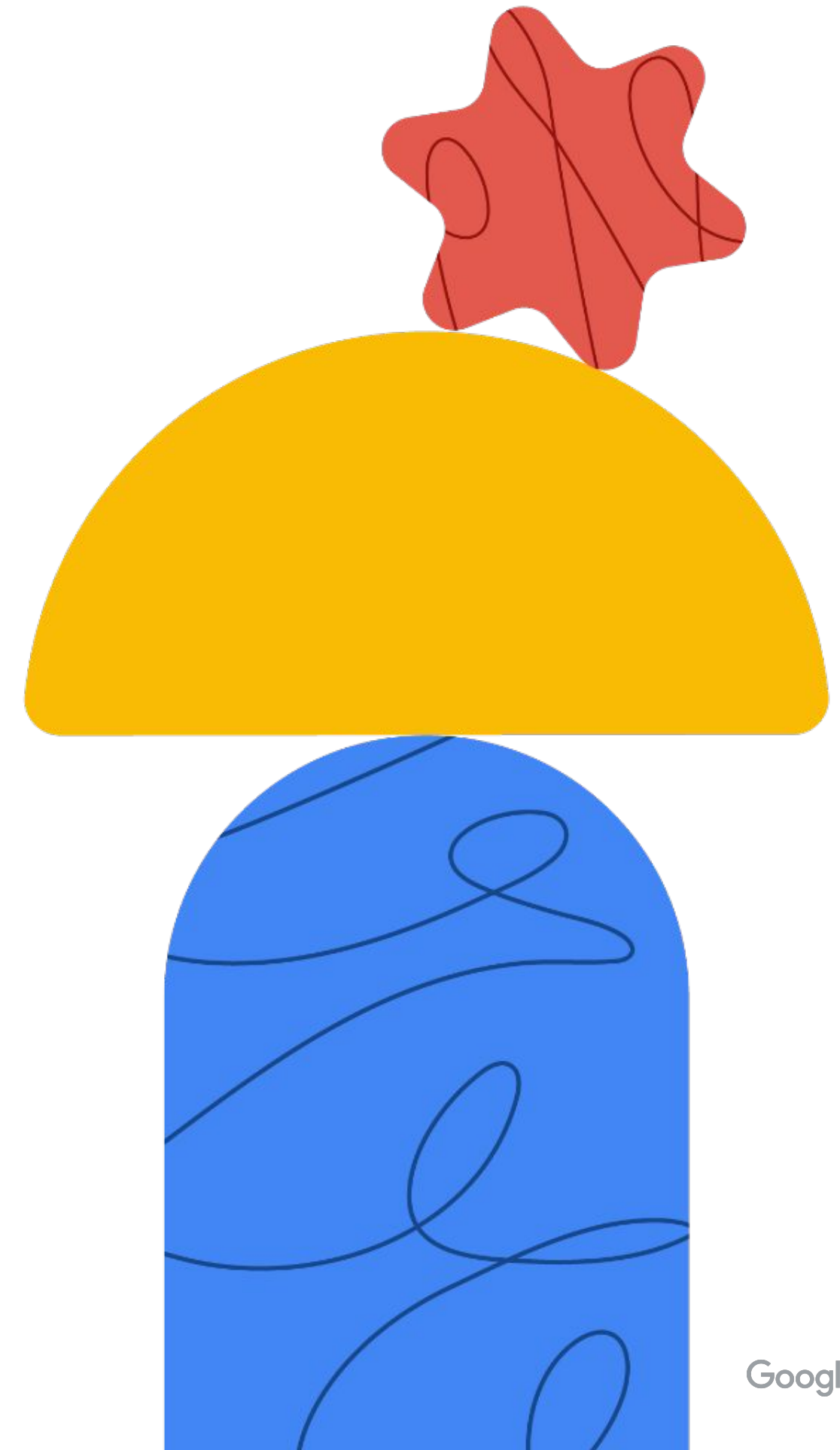
Required application components to be developed and deployed as a single, tightly coupled unit, typically using a single programming language.

Benefits of modern cloud application development

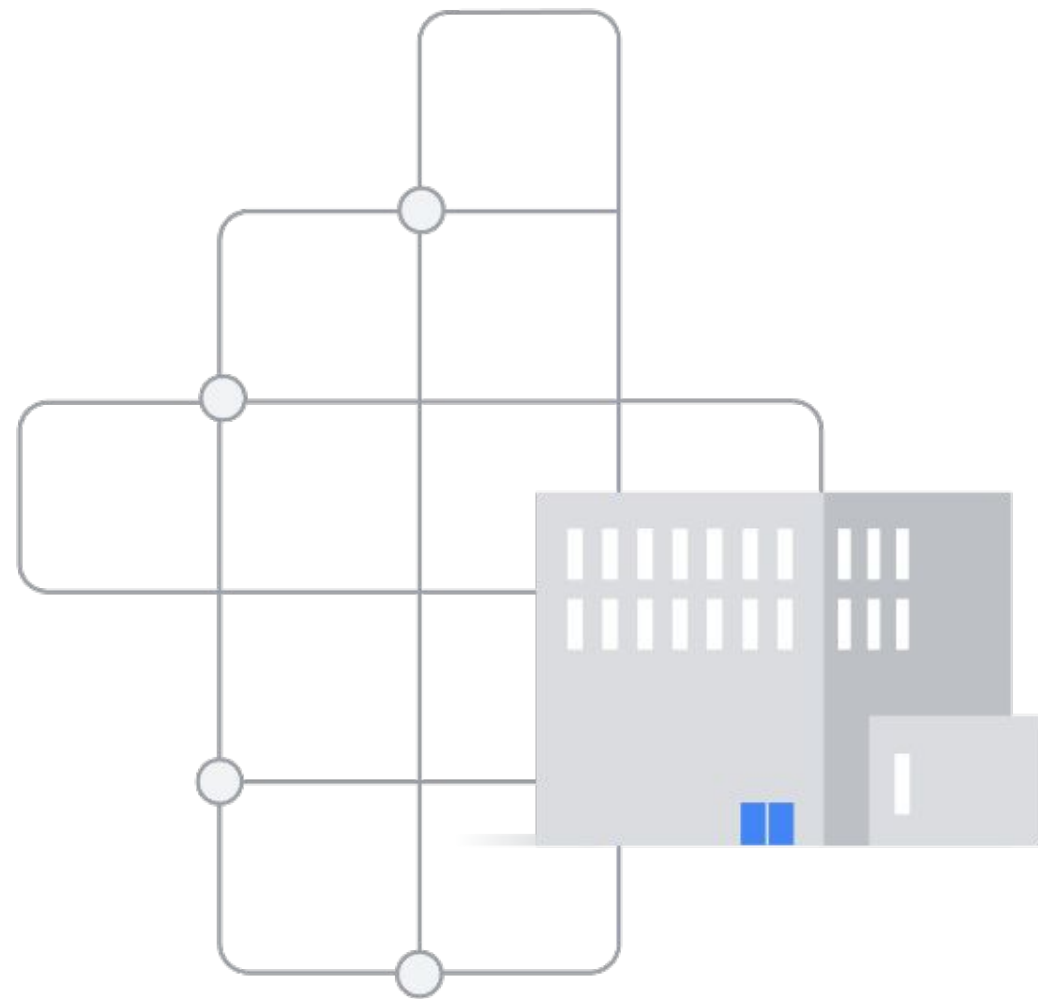


- ✓ Microservices architecture
- ✓ Deployment with managed services
- ✓ Pay-as-you-go pricing
- ✓ Scalability
- ✓ Highly available and resilient
- ✓ Monitoring and management tools

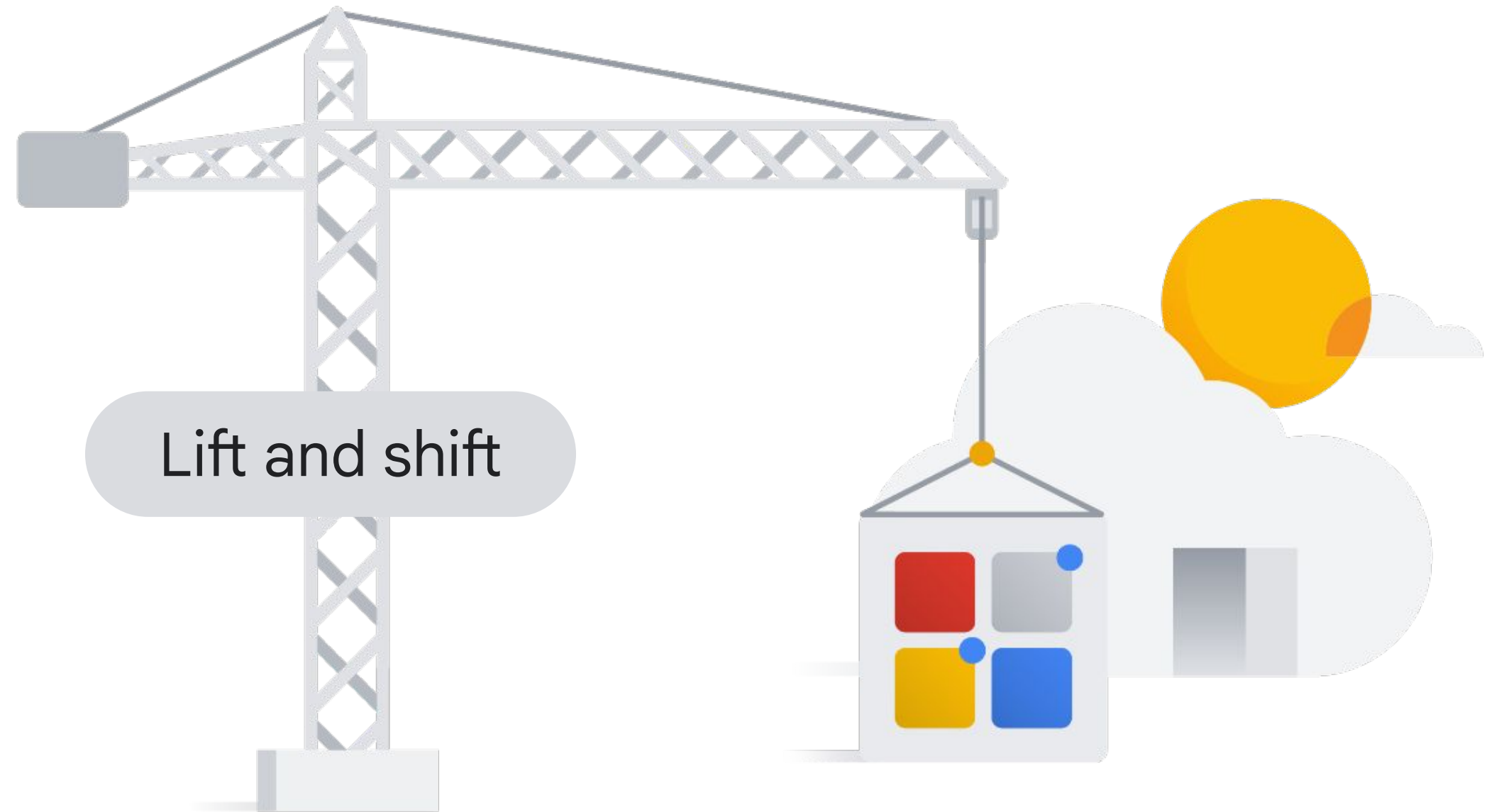
Rehosting legacy applications in the cloud



Sometimes specialized legacy applications aren't compatible with cloud-native applications

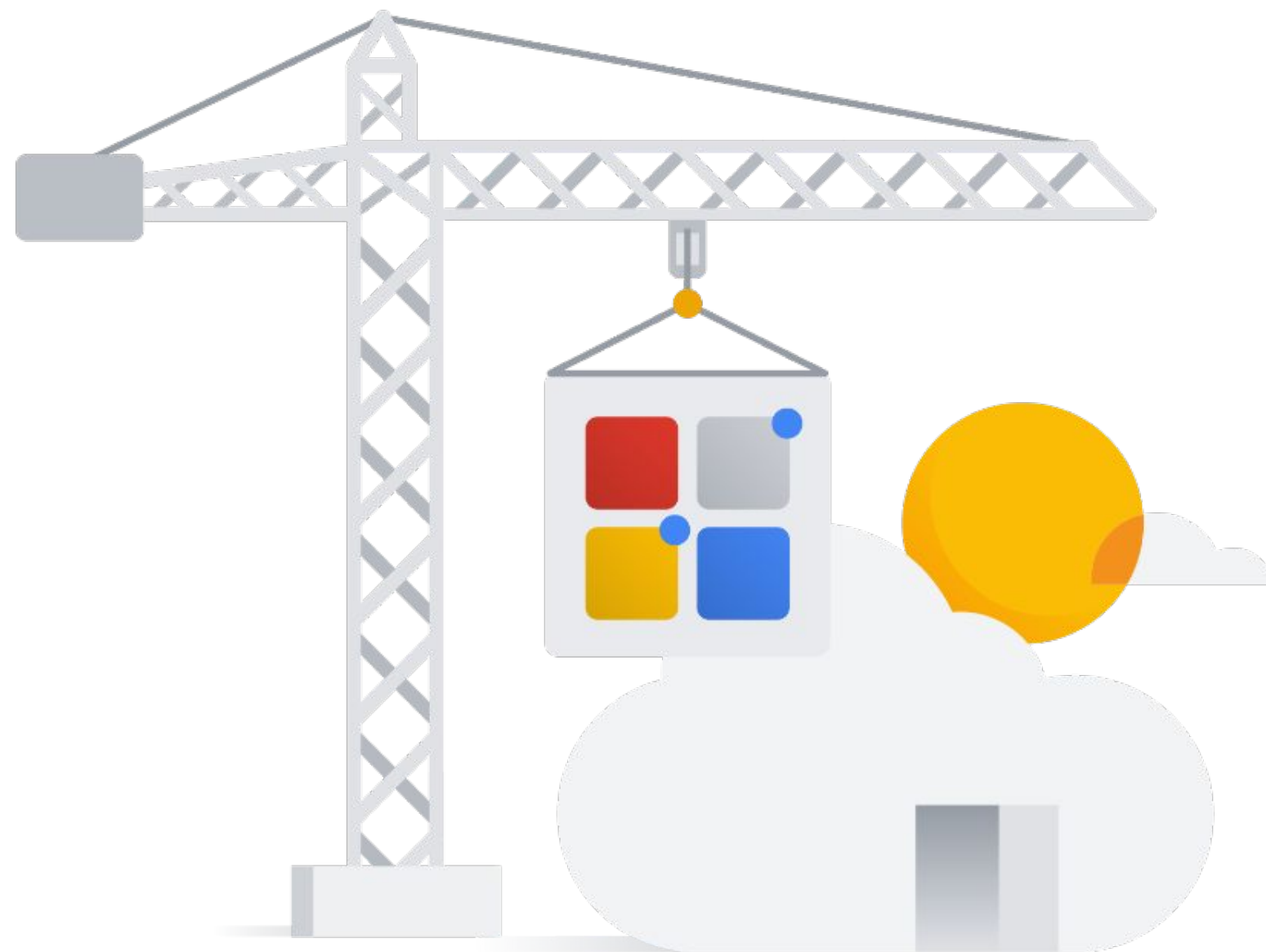


On-premises environment



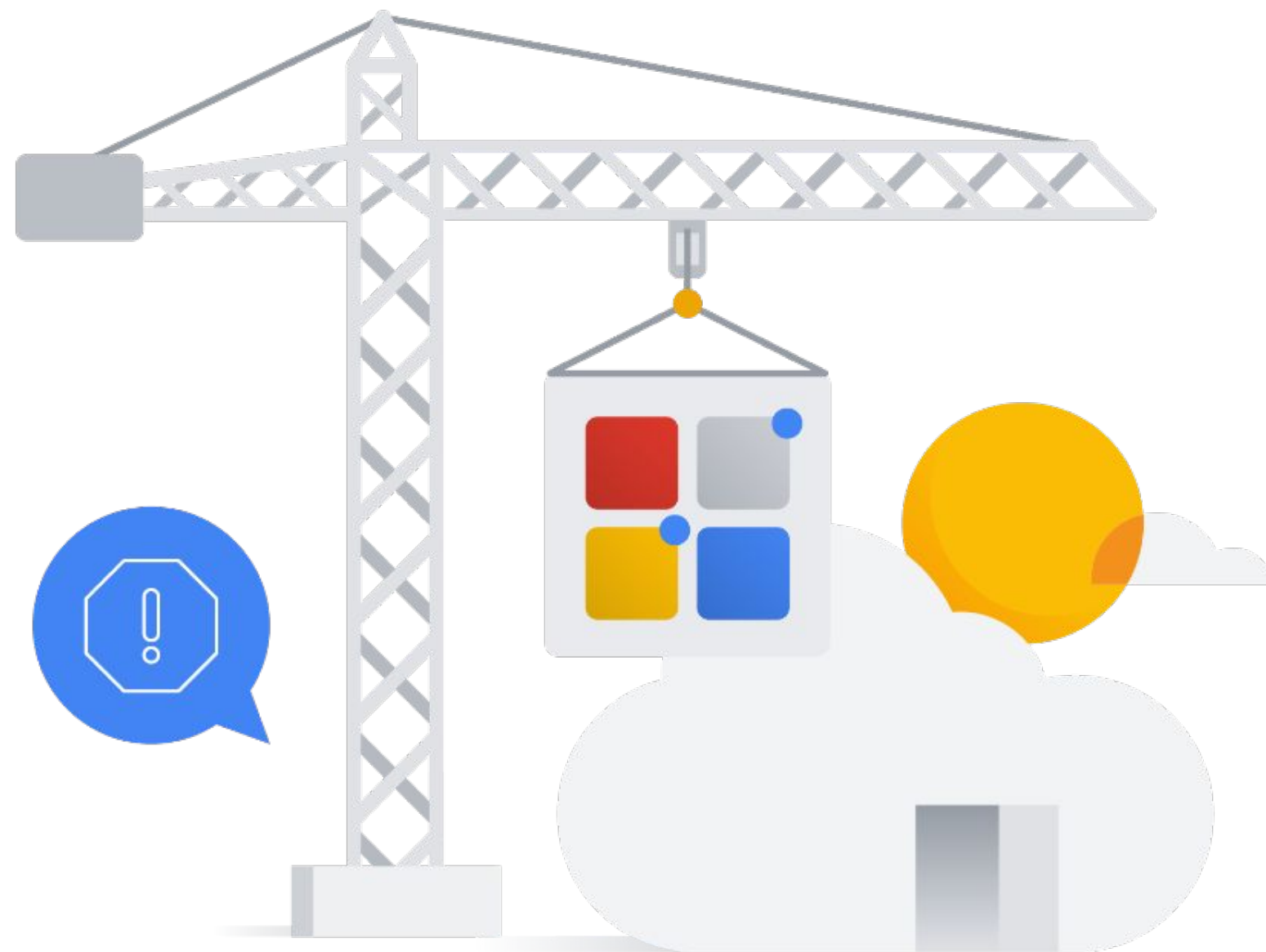
Cloud environment

The benefits of rehosting applications



- ✓ Cost savings
- ✓ Scalability
- ✓ Reliability
- ✓ Security

Potential drawbacks to choosing a rehost migration path for legacy applications



- ! Complexity
- ! Risk
- ! Vendor lock-in

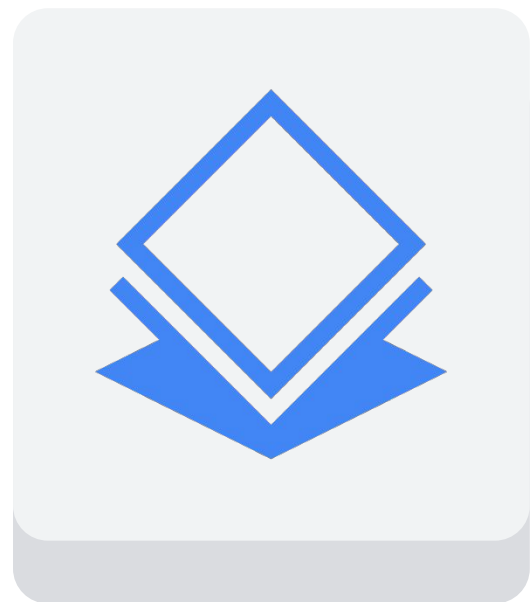
Solutions for rehosting specialized legacy applications



Google Cloud
VMware Engine



Helps migrate existing VMware workloads to the cloud without having to rearchitect the applications or retool operations.



Bare Metal
Solution



Is a fully managed cloud infrastructure solution that lets organizations run their Oracle workloads on dedicated, bare metal servers in the cloud.

Quiz

Question

What phrase refers to when a workload is rehosted without changing anything in the workload's code or architecture.

- A. Move and improve
- B. Refactor and reshape
- C. Reimagine and plan
- D. Lift and shift

Quiz

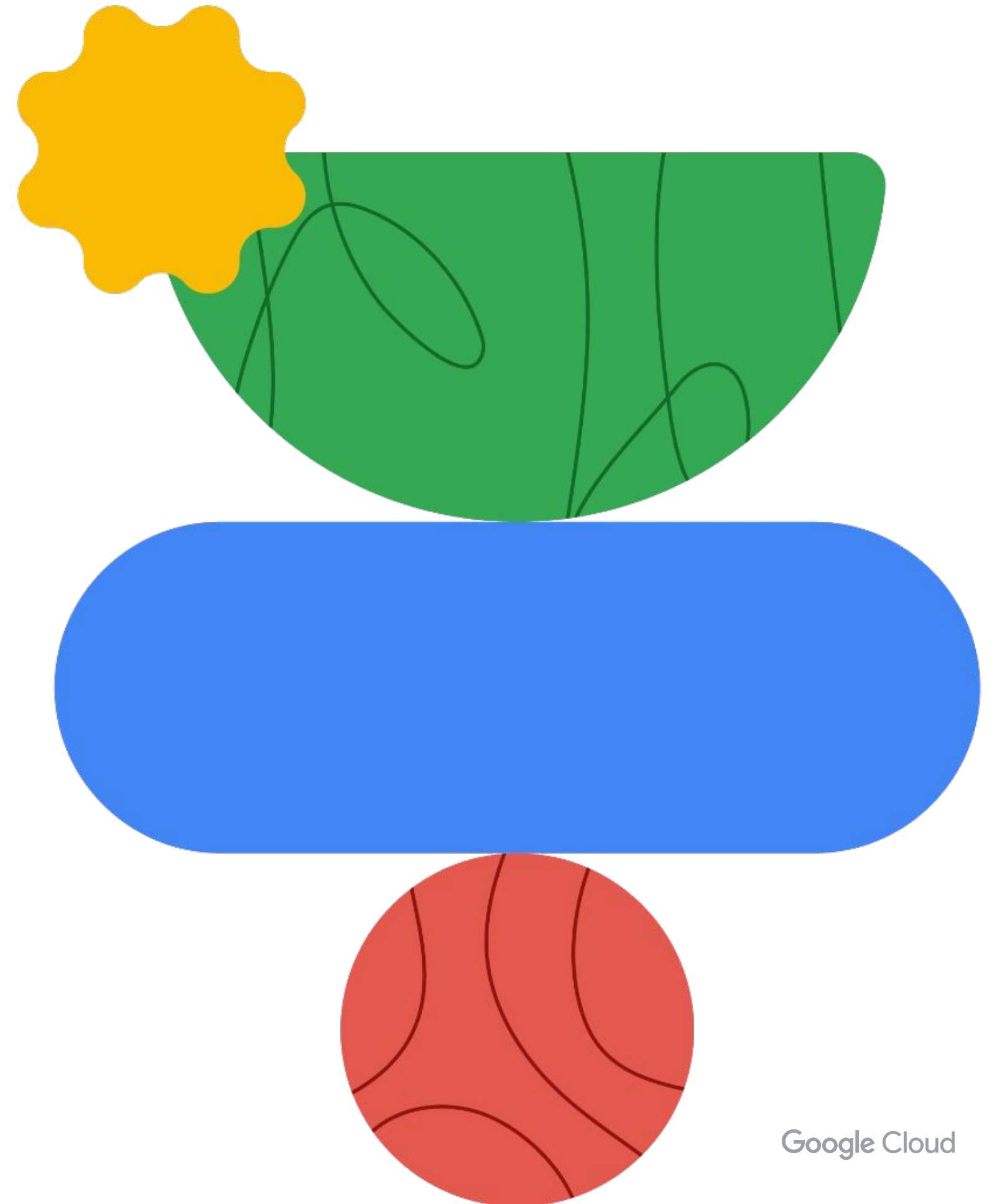
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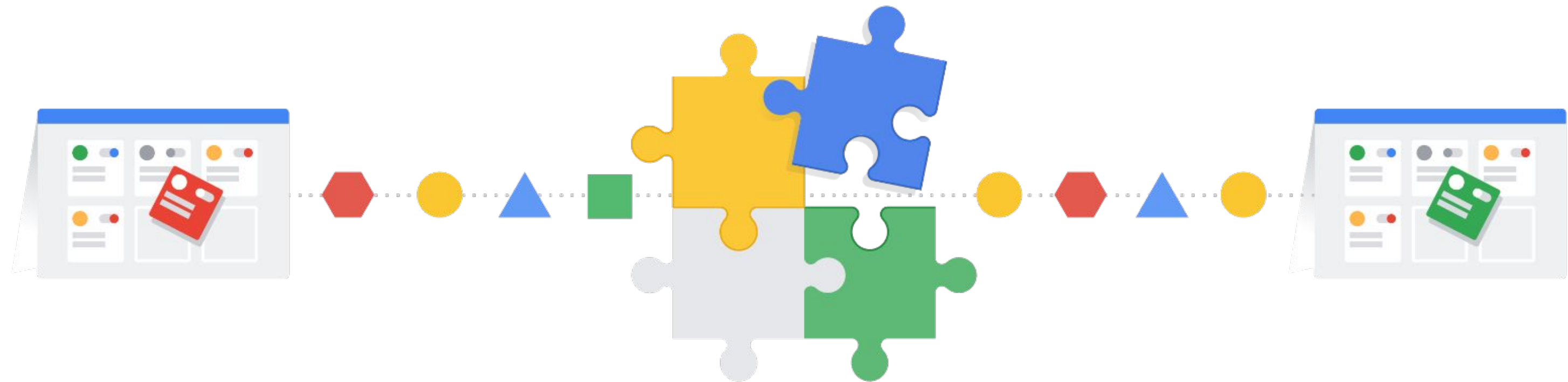
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Application programming interfaces (APIs)

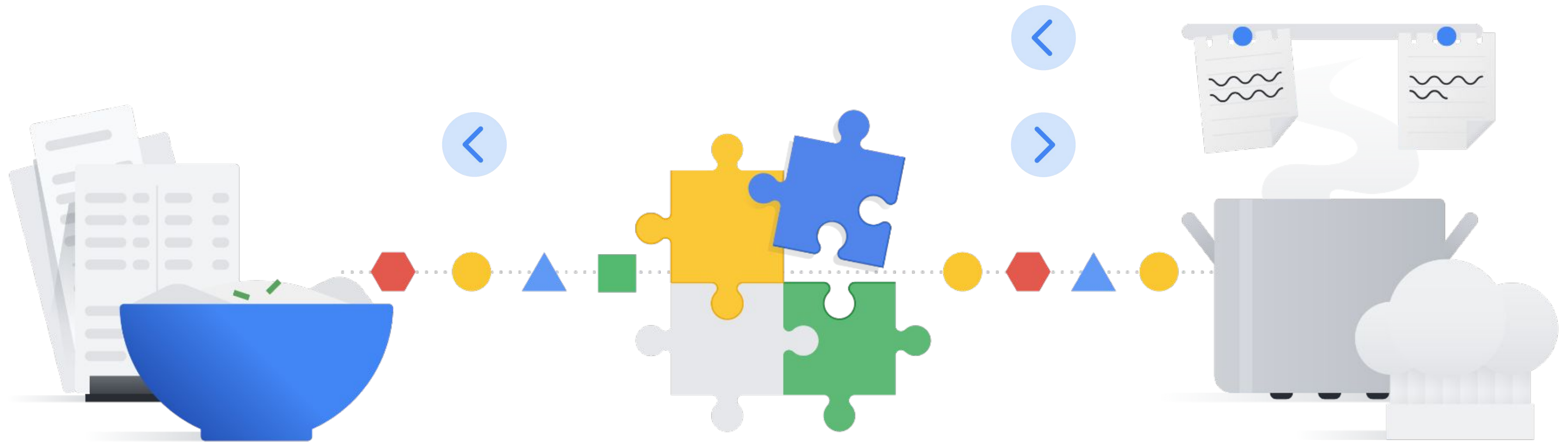


Application programming interfaces (APIs)



A set of instructions that lets different software programs communicate with each other

An API is like a waiter in a restaurant

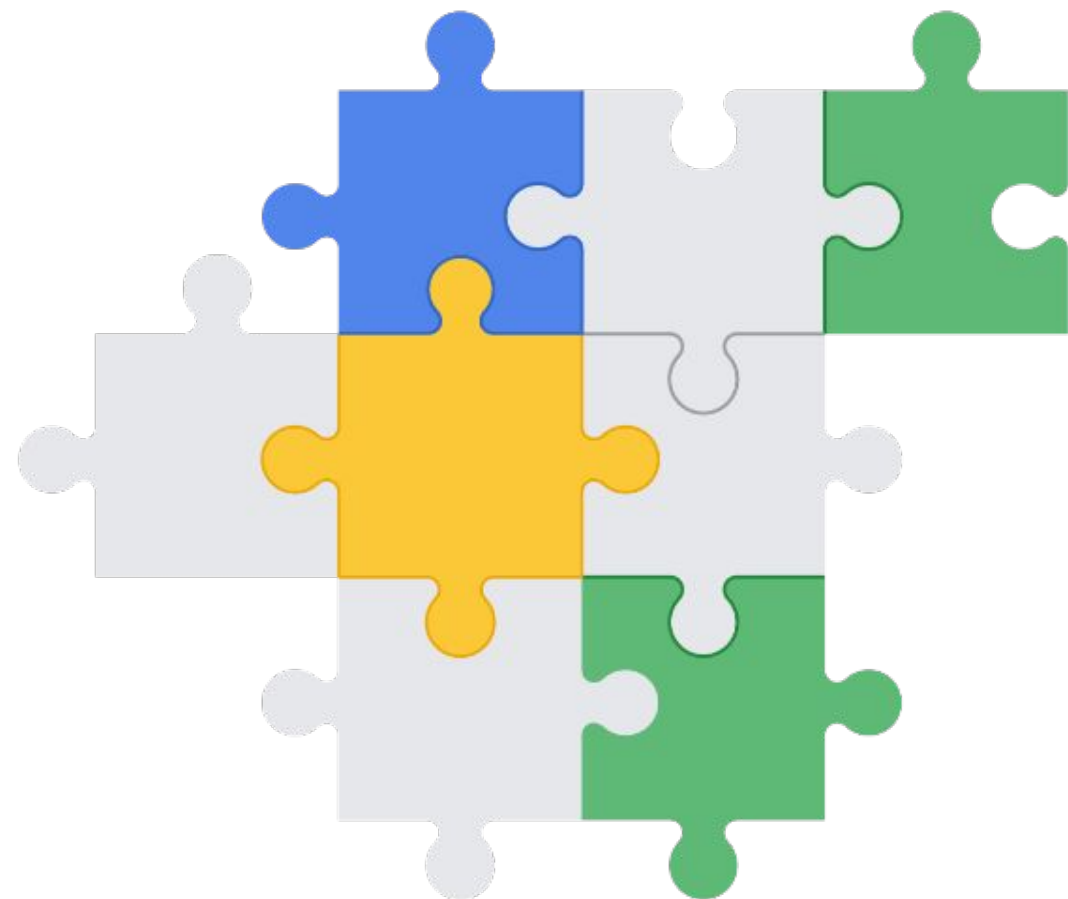


An API takes requests from one software program (the customer).

Communicates with another program (the kitchen).

Returns a response (the food) back to the requesting program (the customer).

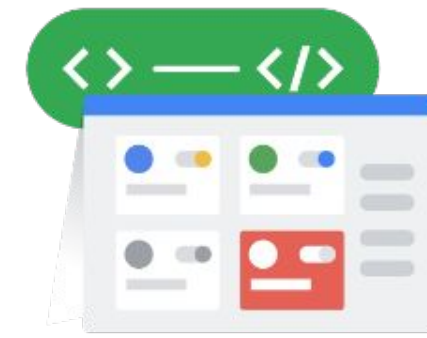
APIs can be used in many different applications



Social media



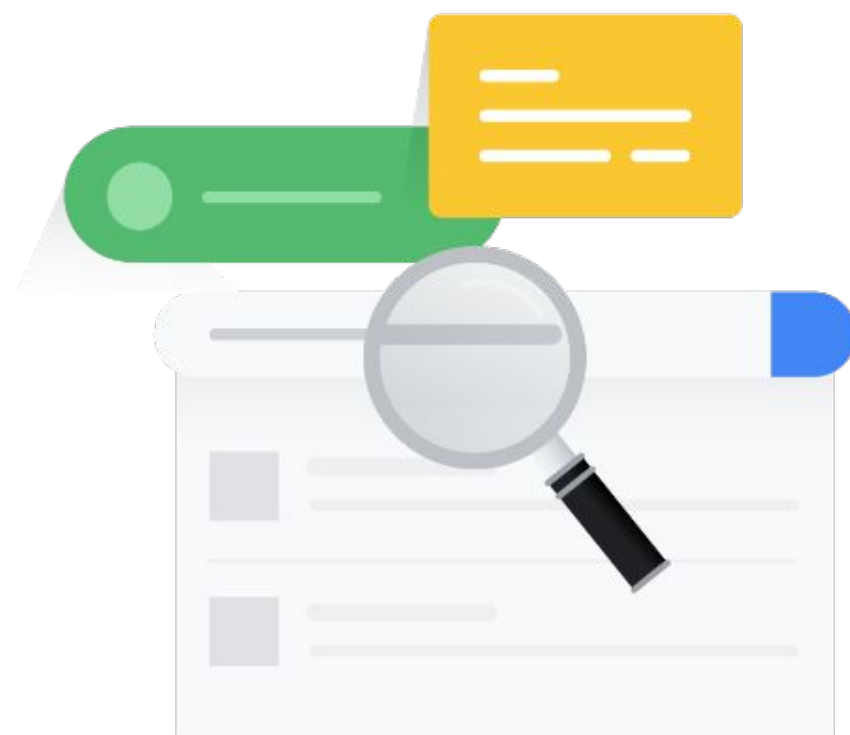
Mobile apps



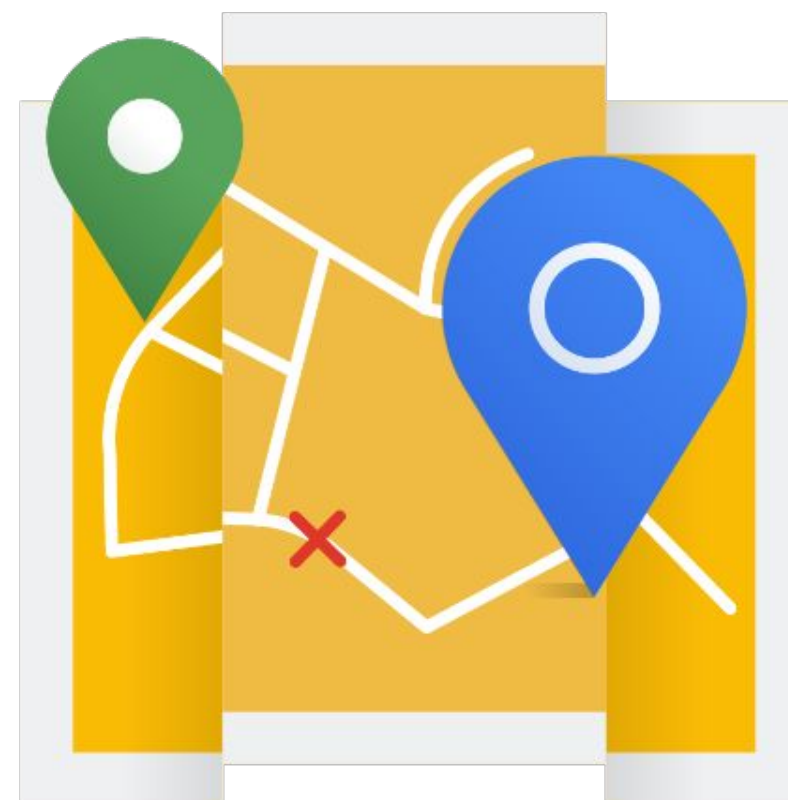
Web services

APIs let developers access functionality and data from other programs without having to write all the code. This saves **time** and **effort**.

Google APIs



APIs that use
the power of
Google to search

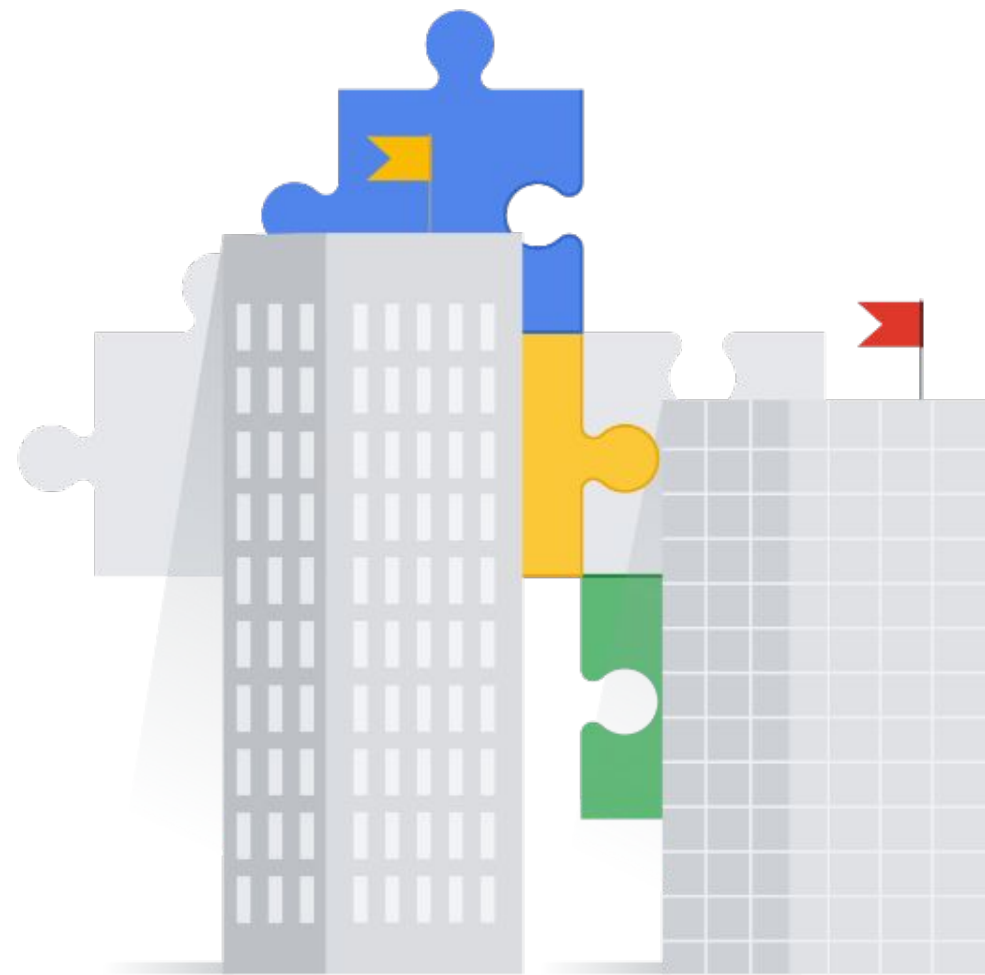


APIs that let
developers access
Google Maps data



APIs that let
developers
translate text

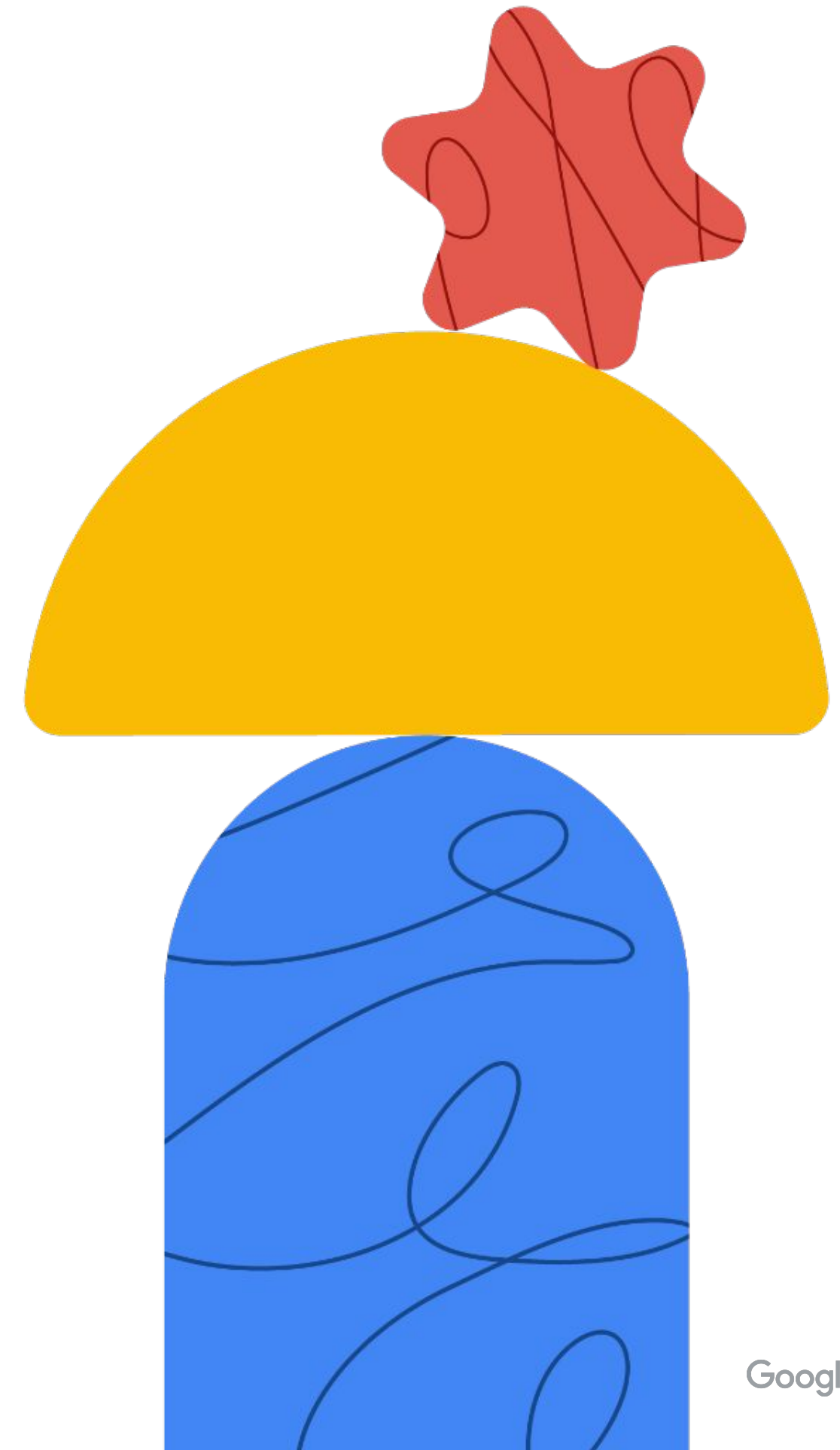
Companies can access an organization's data or services through an API



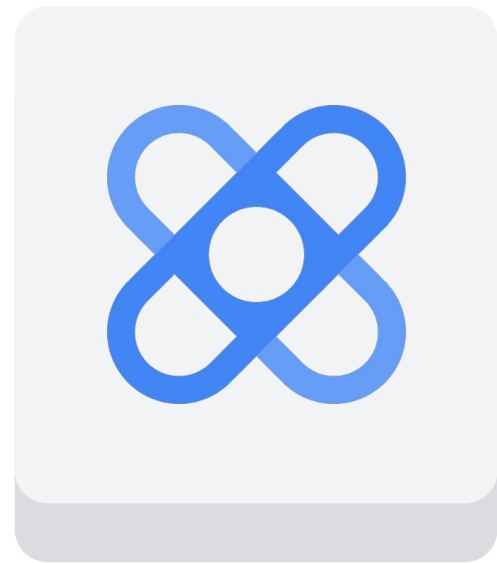
APIs can:

- ✓ Be used to create new products and services
- ✓ Be used to generate new revenue streams
- ✓ Create partnerships

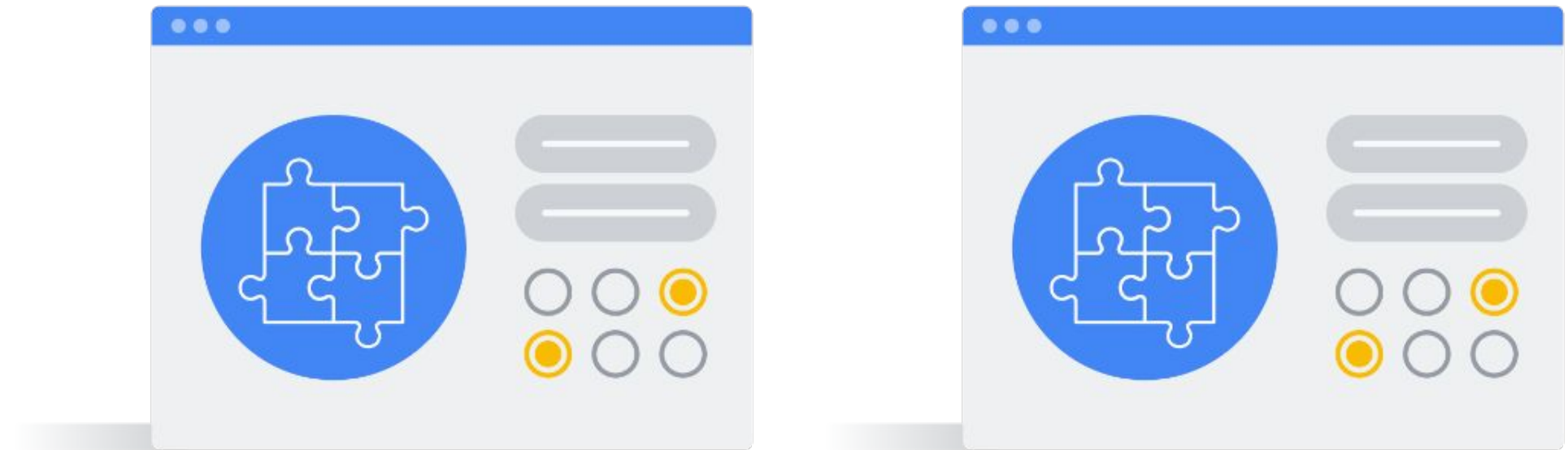
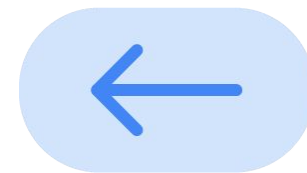
Apigee API Management



Apigee API Management



Apigee API
Management



Google Cloud's API management service to operate
APIs with enhanced scale, security, and automation

Apigee benefits



Apigee API
Management



It helps organizations secure their APIs.



It tracks and analyzes API usage.



It helps with developing and deploying APIs.



It offers API versioning, API documentation,
and even API throttling.

Activity

Identify the Google Cloud product that matches the listed criteria.

Google's serverless platform that allows you to run code without having to manage any infrastructure.

1

Google's virtual machine service that allows you to create and manage virtual machines in the cloud.

2

Helps migrate existing VMware workloads to the cloud without having to rearchitect the applications or retool operations.

3

Google's API management platform that allows you to secure, manage, and publish APIs.

4

Google's managed Kubernetes service that allows you to deploy and manage containerized applications in the cloud.

5

Google's fully managed platform for developing and hosting web applications.

6

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App Engine

Quiz

Question

What open source platform, originally developed by Google, manages containerized workloads and services?

- A. TensorFlow
- B. Go
- C. Kubernetes
- D. Angular

Quiz

Answer

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Quiz

Question

What name is given to an environment where an organization uses more than one public cloud provider as part of its architecture?

- A. Hybrid cloud
- B. Multicloud
- C. Edge cloud
- D. Duocloud

Quiz

Answer

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What term describes a set of instructions that lets different software programs communicate with each other?

- A. Application programming interface
- B. Programming communication link
- C. Network programming interface
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Google Cloud